Attention Guided and Regularised Generative Adversarial Network for High Fidelity Human Face Generation

Abubeker Shehmolo Saide

A Thesis Submitted to the Department of Computer Science and Engineering

College of Electrical Engineering and Computing

Presented in Partial Fulfillment of the Degree of Master's in Computer Science and Engineering (Artificial Intelligence)

Office of Graduate Studies
Adama Science and Technology University

Date: May 28, 2026
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Advisor: Teklu Urgessa (Ph.D., Associate Professor)

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Declaration

I, the undersigned, declare that this thesis entitled **Attention Guided and Regularized Generative Adversarial Network for High Fidelity Human Face Generation** is my own original work and has not been submitted to any other university or institution for the award of any degree or similar purpose. All sources of materials and references used in this thesis have been duly acknowledged through proper citations.

Abubeker Shehmolo Saide		
Name of Student	Signature	Date

Recommendation of Advisors

I, the major advisor of this Thesis, hereby certify that I have closely advised the student while developing this Thesis and read the draft Thesis entitled **Attention guided and regularised Generative Adversarial Network for high fidelity human face generation** prepared under my guidance by Abubeker Shehmolo Saide. Therefore, I recommend the submission of the Thesis to the department for further review and evaluation.

Dr. Teklu Urgessa	____________	____________
____________	Signature	Date
Major Advisor		
Co-advisor	____________	____________
____________	Signature	Date

Approval sheet

I, the major advisor of this Thesis, hereby certify that the recommendation and suggestion given by the research review committee are appropriately incorporated into the final Thesis entitled **Attention guided and regularised Generative Adversarial Network for high fidelity human face generation** by Abubeker Shehmolo Saide.

Dr. Teklu Urgessa		
Major Advisor	Signature	Date
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Approval of Board of Reviewers

We, the undersigned, members of the Board of Examiners of the thesis open defense by Abubeker Shehmolo Saide have read and evaluated the thesis entitled **Attention guided and regularised Generative Adversarial Network for high fidelity human face generation** and examined the candidate's understanding of the research. This is, therefore, to certify that the thesis is accepted as fulfilling the requirements for the degree of Master of Science in Artificial Intelligence.

Chairperson	Signature	Date
Reviewer 1	Signature	Date
Asrat Mulatu (Ph.D)		08-Jun-2026
Reviewer 2	Signature	Date
SIntayehu H. (PhD)		11-June-2026

Final approval and acceptance of this thesis is contingent upon submission of its final copy to the Office of Postgraduate Studies (OPGS) through the Department Graduate Council (DGC) and College Graduate Committee (SGC).

Department Head	Signature	Date
College Dean	Signature	Date
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List of Acronyms

AGRDCGAN	Attention-Guided and regularised Deep Convolutional Generative Adversarial Network
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CUDA	Compute Unified Device Architecture
CW-SSIM	Complex Wavelet Structural Similarity Index
DCGAN	Deep Convolutional Generative Adversarial Network
EMA	Exponential Moving Average
FFHQ	Flickr-Faces-HQ
FID	Fréchet Inception Distance
FPR	False Positive Rate
GAN	Generative Adversarial Network
GB	Gigabyte
GPU	Graphics Processing Unit
IDE	Integrated Development Environment
L2	Level-2 Regularization
LFW	Labeled Faces in the Wild
LMHCSA	Local Multi-Head Channel Self-Attention
MB	Megabyte
SN-GAN	Spectrally Normalized Generative Adversarial Network
PIL	Python Imaging Library
PyTorch	An open-source deep learning library based on Python
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
SAGAN	Self-Attention Generative Adversarial Network
STL-10	Stanford Tiny Images Dataset

Abstract

Face generation is a difficult problem in computer vision because of the complex structure of faces, fine-grained textures, and long-range spatial relationships. In recent years, deep generative modeling has made great strides, especially in the field of Generative Adversarial Networks, which have enhanced the quality of synthesized image generation. In particular, face synthesis has been successfully done using Deep Convolutional Generative Adversarial Networks, but traditional DCGANs still exhibit problems with training instabilities, mode collapses, and limited ability to retain detailed facial structures and global consistency. This thesis proposes an Attention-Guided and Regularized Deep Convolutional Generative Adversarial Network for high-fidelity human face generation. The proposed model is based on DCGAN with the incorporation of Local Multi-Head Channel Self-Attention modules to enhance the ability of capturing long-range feature dependency and structural consistency in facial regions. Furthermore, spectral normalization and weight decay are incorporated to stabilize adversarial training, prevent mode collapse, and accelerate convergence. A subset of the Flickr-Faces-HQ dataset was preprocessed and augmented to train the model. The performance of the proposed model was tested in terms of Fréchet Inception Distance, Complex Wavelet Structural Similarity Index and qualitative visual tests for the assessment of image realism, structural consistency, texture similarity and artefact reduction. Additionally, experiments on downstream face verification were performed on Labeled Faces in the Wild dataset to assess the identity preservation capability. Experimental results show that the proposed Attention-Guided and Regularized Deep Convolutional Generative Adversarial Network outperformed the conventional Deep Convolutional Generative Adversarial Network model, achieving a best validation Fréchet Inception Distance score of 30.93 and a test Fréchet Inception Distance score of 35.73 while maintaining high structural consistency. In downstream evaluation, the model achieved a face verification accuracy of 99.80% and a ROC-AUC score of 1.0000, demonstrating strong identity preservation capability and discriminative performance. The findings indicate that integrating attention mechanisms and regularization techniques into DCGAN significantly improves image realism, training stability, and the effectiveness of generated images for biometric applications.

Keywords: Deep Convolutional Generative Adversarial Network, Generative Adversarial Network, Face Generation, Self-Attention, Image Synthesis, Fréchet Inception Distance, Complex Wavelet Structural Similarity Index, Deep Learning, Biometric Recognition.

CHAPTER ONE

INTRODUCTION

1.1 Background

The human face is one of the most informative biological structures and has a fundamental role in the recognition of identity, expression of emotions, communication in society and interaction in behavior. Facial appearance is a complex visual pattern that is constructed from the eyes, nose, mouth, eyebrows, jawline, skin texture and spatial relationships between the facial features. Face recognition with humans is successful in a variety of situations including different lighting, different poses, occlusion, and facial expressions. This complexity has generated great interest in the field of facial image analysis in computer vision, pattern recognition and artificial intelligence (Zhao et al., 2003).

However, recent progress in Deep Learning has revolutionized Computer Vision and now computers are able to accurately classify images, identify objects, segment images, and generate images (LeCun et al., 2015). In particular, generative modeling has received significant interest as it learns the distribution of the data and produces new samples that are similar to real data. It is widely used in data enhancement, simulation, privacy-preserving learning, content generation, and representation learning (Kingma & Welling, 2014).

Generative adversarial networks are a significant breakthrough in generative modelling. They are based on a generator and a discriminator, which are trained in an adversarial fashion: The generator generates synthetic samples and the discriminator judges the authenticity of the samples produced. This competition is a way the generator iteratively gets closer to the actual data distribution and generates realistic data samples (Goodfellow et al., 2014).

The image synthesis applications of GAN have been demonstrated in human face generation, a difficult task because it is characterized by the complexity of the structures, the variability of the texture and the sensitivity to the identity of the face. In order to perform face synthesis, symmetry, texture consistency and geometric relations between the components of the face must be maintained (Karras et al., 2019).

Deep Convolutional Generative Adversarial Network was introduced to enhance the performance in adopting convolutional neural networks to the adversarial network. It does the same as replacing fully connected layers with convolutional layers, which in turn facilitates good learning of spatial features using the strided convolutions, batch normalization, and better activation functions (ReLU, Leaky ReLU). These improvements will increase stability and image quality (Radford et al., 2016).

On the benchmark datasets, Celebfaces Attributes Dataset and Flickr-Faces-HQ, DCGAN is able to generate realistic facial images with various identities and expressions (Liu et al., 2015; Karras et al., 2019). The synthetic images are utilized in biometric systems, data enhancement, privacy-preserving identity generation and virtual avatar applications.

Although its success, DCGAN is not good at capturing the long-range dependencies between facial regions, because convolutional operations have limited receptive fields and can not model the global dependency like facial symmetry and consistency on the structure of the face (Zhang et al., 2019).

It also has a trouble with fine-grained details like eyelashes, teeth alignment, strands of hair and texture of skin, which can lead to blurry, unrealistic results. Further, the adversarial training is not stable and can suffer from mode collapse, vanishing gradients, non-convergence and discriminator dominance (Salimans et al., 2016).

In order to overcome these drawbacks, attention mechanisms have been added. They are useful for focusing on the most salient parts of a model, and can improve the local detail fidelity and global structural consistency, allowing them to highlight key facial areas, like the eyes, nose or mouth (Vaswani et al., 2017; Zhang et al., 2019).

Other methods for regularization and robust training and generalization, including spectral normalization, gradient penalty, feature matching and weight decay (Miyato et al., 2018), are also employed.

There must be both quantitative and qualitative approaches to evaluation of generative models. The similarity between real and generated distributions is measured using deep feature representations through Fréchet Inception Distance, the lower the value the better the generated quality (Heusel et al., 2017). Wang et al. (2004) proposed two structural similarity index measures: perceptual similarity index measure and complex wavelet structural similarity measure to assess perceptual and structural consistency between images.

Evaluation of downstream is also significant to determine whether synthetic images preserve the identity information. Use classification accuracy and receiver operating characteristic area under curve measure identity preservation capabilities to perform face verification tasks.

In recent research, it has been found that representation learning and data augmentation are enhanced by high-quality synthetic face datasets which have been found to be highly dependent on realism, diversity and structural accuracy (Shorten & Khoshgoftaar, 2019).

Inspired by such challenges, in this thesis, we present Attention Guided and regularised DCGAN to generate high-quality face images. The model incorporates attention mechanisms, regularization techniques, and other modifications to learn features more effectively, reduce the instability of the learning process, and improve realism. Regularization helps with convergence stability, and attention helps to capture long-range dependencies.

The proposed model is designed to create face images that are structurally consistent, have finer details, and more coherent identity. The model is validated using the Fréchet Inception Distance, the complex wavelet structural similarity measure as well as by visual evaluation. Moreover, experiments for face verification are performed with the Labeled Faces in the data.

This thesis is beneficial for generative modeling for improving realism, diversity and stability of face image synthesis in biometric systems, forensic analysis, generation of face images while maintaining privacy and improving deep learning.

1.2 Motivation of Study

Human face generation has become an important research area in computer vision due to the increasing demand for realistic synthetic face images in applications such as entertainment, virtual reality, augmented reality, biometric systems, privacy-preserving technologies, and data augmentation. The ability to generate high-quality synthetic faces can reduce dependence on large-scale real-world facial datasets while enabling diverse and controllable data synthesis for various artificial intelligence applications.

Recent advances in deep learning-based generative models have significantly improved image synthesis performance. Approaches such as GAN-based architectures, style-based models, attention-guided frameworks, and diffusion models have demonstrated strong capabilities in generating realistic facial images with improved semantic consistency and visual fidelity. However, despite these advancements, high-fidelity human face generation remains a challenging problem due to the complexity of facial structures, variations in pose and expression, and the need to preserve both local texture details and global structural coherence.

Existing methods still suffer from several limitations. Traditional GAN-based approaches, such as DCGAN, often experience training instability, mode collapse, and difficulties in capturing fine-grained facial details. Although attention-based models such as Self-Attention GANs improve global feature dependency modeling, they typically introduce high computational cost and memory consumption. Similarly, style-based and text-guided generation methods improve controllability and realism but may suffer from architectural complexity or limited adversarial stability. Furthermore, diffusion-based models achieve high-quality image synthesis but are computationally expensive and less efficient for large-scale generation tasks.

To address these challenges, recent studies have explored the integration of attention mechanisms and regularization techniques to enhance feature representation and improve training stability. Attention modules help models focus on both local and global facial relationships, while regularization methods such as spectral normalization and L2 regularization improve convergence behavior and stabilize adversarial training.

Therefore, this thesis is motivated by the need to develop a unified and efficient generative model that integrates a LMHCSA mechanism with regularization strategies. The proposed approach aims to improve training stability, enhance structural consistency, and generate high-fidelity human face images with better texture detail preservation and perceptual realism while maintaining computational efficiency.

1.3 Statement of the Problem

Generating high-fidelity human face images remains a significant challenge in computer vision due to the inherent complexity of facial structures, fine-grained textures, and long-range spatial dependencies. Although Deep Convolutional Generative Adversarial Networks have demonstrated promising results in face synthesis, conventional DCGAN architectures suffer from three persistent limitations: training instability caused by imbalanced adversarial dynamics, mode collapse resulting in limited diversity of generated faces, and inadequate modeling of long-range feature dependencies due to the restricted receptive fields of convolutional operations. These limitations collectively result in generated images with structural inconsistencies, checkerboard artifacts, blurry facial features, and weak global coherence.

Attention mechanisms, particularly self-attention, have been proposed to address the dependency modeling problem. However, existing self-attention approaches such as SAGAN introduce quadratic computational complexity, making them impractical for resource-constrained environments. Furthermore, experimental evidence indicates that integrating attention mechanisms without complementary regularization strategies can destabilize adversarial training rather than improve it, a critical interaction that remains insufficiently explored in the literature.

Regularization techniques such as spectral normalization and L2 weight decay have independently demonstrated improvements in training stability; however, their principled integration with lightweight attention mechanisms within a unified DCGAN framework has not been systematically investigated. Existing methods either focus on computationally intensive architectures such as StyleGAN2 and diffusion models, which are often inaccessible in constrained environments, or address individual limitations in isolation without considering their combined effect on generation quality and training stability.

Therefore, there exists a clear need for a computationally efficient and unified generative framework that integrates lightweight attention mechanisms with complementary regularization strategies to simultaneously improve training stability, structural consistency, and perceptual realism of generated human face images within resource-constrained settings.

1.4 Research Questions

This thesis will answer the following research questions:

Research Question 1: To what extent do the spatial coherence and fidelity of generated human faces increase when attention processes are incorporated into the DCGAN architecture?

Research Question 2:When compared to baseline DCGAN models, how well can particular regularization techniques reduce training instabilities and semantic mismatches?

Research Question 3: To what extent do evaluation metrics such as Fréchet Inception Distance and Structural Similarity Index reflect the performance improvements achieved by attention-guided and regularized DCGAN models in generating diverse and realistic human faces?

This thesis aims to design and test an Attention Guided and regularised DCGAN for high fidelity human face generation.

1.5 Research Objectives

1.5.1 General Objective

The goal of this thesis is to design and evaluate an Attention Guided and regularised DCGAN to synthesize 2D human face images.

1.5.2 Specific Objectives

- To design and implement an Attention-Guided and Regularized Deep Convolutional Generative Adversarial Network by integrating Local Multi-Head Channel Self-Attention into the DCGAN architecture.
- To enhance adversarial training stability through the integration of spectral normalization and L2 regularization techniques.
- To generate high-quality and structurally consistent 128×128 human face images using the proposed AGRDCGAN model trained on the FFHQ dataset.
- To evaluate the quality and structural consistency of the generated face images using Frechet Inception Distance, Complex Wavelet Structural Similarity Index, and qualitative visual assessment.
- To compare the performance of the proposed AGRDCGAN model with the baseline DCGAN in terms of image realism, structural consistency, and training stability.
- To assess the effectiveness of the proposed attention and regularization mechanisms through ablation studies and downstream face verification experiments.

1.6 Scope of the Study

The proposed AGRDCGAN model is developed and evaluated within clearly defined boundaries to ensure research focus, reproducibility, and feasibility within a Master's thesis framework. The study utilizes a 10,000-image subset of the FFHQ dataset, selected using a fixed random seed (42) to ensure experimental reproducibility. FFHQ was chosen due to its high image quality and diversity in age, gender, ethnicity, facial expressions, poses, illumination conditions, and accessories, making it suitable for generative face modeling research. Identity preservation is further evaluated using the LFW dataset through a downstream face verification task.

This research focuses on unconditional 2D human face generation at a resolution of 128×128 pixels. The primary objective is to improve image quality, training stability, and structural consistency over the conventional DCGAN baseline through the integration of attention mechanisms and regularization techniques. AGRDCGAN is intentionally designed as a lightweight and computationally efficient enhancement of DCGAN rather than a replacement for resource-intensive architectures such as StyleGAN2 or diffusion-based models. The study aims to demonstrate that targeted architectural modifications can produce meaningful improvements in face generation quality while remaining practical for single-GPU environments.

Model performance is evaluated using Fréchet Inception Distance and Complex Wavelet Structural Similarity Index , while a downstream face verification task on LFW assesses identity preservation. In addition, a systematic ablation study is conducted to quantify the contribution of each architectural component, supported by qualitative visual inspection of generated samples.

The scope of this research excludes 3D face synthesis, video generation, face editing, real-time deployment, mobile and edge-device optimization, conditional face generation, cross-dataset generalization beyond LFW, higher-resolution image generation, Ethiopian-specific face datasets, formal ethical and legal deployment frameworks, and direct comparisons with StyleGAN2, diffusion-based, and transformer-based generators under identical training conditions due to computational resource constraints. These boundaries establish a focused, rigorous, and achievable thesis contribution while providing clear directions for future research.

1.7 Significance of the Study

This research makes the following specific scientific contributions to the fields of computer vision and generative modeling:

Scientific Contribution 1: Systematic Integration of Attention and Regularization

This thesis provides a systematic empirical study demonstrating that LMHCSA attention and regularization techniques are complementary in adversarial face generation. The ablation study reveals a critical non-trivial finding: attention mechanisms alone without regularization degrade both FID and CW-SSIM performance, while their principled combination yields the best results. This finding directly informs future GAN architectural design decisions.

Scientific Contribution 2: Computational Efficiency Analysis

The LMHCSA module reduces attention computational complexity from $O(H^2W^2C)$ to approximately O(HWC), achieving roughly a 1000× reduction compared to global self-attention while still delivering meaningful improvements in generation quality. This demonstrates that efficient, lightweight attention mechanisms can be effectively integrated into constrained generative architectures.

Scientific Contribution 3: Downstream Utility Validation

The thesis provides empirical evidence that AGRDCGAN-generated images improve face verification accuracy from 99.60% to 99.80% over vanilla DCGAN augmentation on LFW, demonstrating the practical value of higher-quality synthetic face images for biometric data augmentation applications.

Scientific Contribution 4: Reproducible Experimental Framework

A fully documented, reproducible experimental pipeline with fixed random seeds, detailed hyperparameter configurations, systematic ablation analysis, and complete implementation code is provided, supporting future research reproducibility and extension.

Broader Significance:

- **Synthetic Face Generation for Digital Applications:** High-quality synthetic face images can be used in virtual environments, gaming platforms, and entertainment systems to minimize the need for manual content creation and boost user experience.

- **Advancement of Biometric and Security Systems:** Improved face generation techniques can create diverse, high-quality datasets for training and evaluating facial recognition systems, improving the accuracy, robustness, and fairness of biometric applications.

- **Support for Data Augmentation:**By creating realistic synthetic face images, data scarcity and class imbalance problems can be mitigated, leading to better performance and generalization of face analysis models.

- **Efficient and Scalable Deep Learning Solutions:** The research explores ways to enhance the performance of generative models without incurring excessive computational costs and to support the synthesis of high-quality faces on various platforms and applications, providing efficient and scalable solutions.

CHAPTER TWO

LITERATURE REVIEW

2.1 Overview

Computer vision is a key field of Artificial Intelligence that enables computers to analyze and interpret visual information from images and videos. Advances in deep learning have expanded computer vision beyond traditional discriminative tasks such as image classification and facial recognition toward generative modeling, where systems synthesize realistic data that follow the distribution of real-world observations (LeCun et al., 2015). Human face generation remains particularly challenging due to the complexity of facial structures, identity-specific features, pose variations, illumination changes, and fine-grained textures. Successful face synthesis requires preserving both global facial structure and local facial characteristics while maintaining realism and identity consistency (Karras et al., 2019).

The introduction of Generative Adversarial Networks by Goodfellow et al. (2014) marked a major breakthrough in generative modeling through an adversarial framework consisting of a generator and discriminator. Building on this foundation, Radford et al. (2016) proposed Deep Convolutional GANs, which improved training stability and image quality through convolutional architectures, batch normalization, and stable activation functions. Due to their simplicity and computational efficiency, DCGANs remain relevant for practical face generation applications. Recent studies such as Lv et al. (2024) and Manimegala et al. (2024) demonstrated that lightweight DCGAN variants continue to achieve meaningful improvements in face synthesis and recognition tasks. Nevertheless, conventional DCGANs still suffer from limitations including restricted long-range dependency modeling, structural inconsistencies, mode collapse, and gradient instability (Salimans et al., 2016). A systematic review by Cobbinah et al. (2025) further reported that combining architectural modifications with regularization strategies consistently outperforms either approach independently.

To address adversarial training instability, several regularization techniques have been proposed. Spectral normalization improves gradient stability by constraining discriminator weight matrices (Miyato et al., 2018), while L2 regularization reduces overfitting and enhances generalization (Krogh & Hertz, 1992). Kurach et al. (2019) demonstrated that regularization significantly influences the effectiveness of architectural components, particularly attention mechanisms, which may amplify training instability when not properly stabilized.

Attention mechanisms have become increasingly important for modeling long-range feature dependencies. Self-Attention GAN (SAGAN) incorporated global self-attention into GANs and improved structural consistency in generated images (Zhang et al., 2019). However, global self-attention introduces substantial computational overhead. To improve efficiency, Liu et al. (2021) proposed window-based local attention, while Pecoraro et al. (2022) introduced LMHCSA for efficient channel-wise dependency modeling. Despite these advances, the combined integration of lightweight attention and complementary regularization strategies within a unified adversarial framework remains insufficiently explored.

StyleGAN-based architectures currently represent the state of the art in high-fidelity face generation. StyleGAN, StyleGAN2, and StyleGAN3 significantly improved image quality, controllability, and identity preservation through style-based latent representations (Karras et al., 2019; 2020; 2021). Recent studies, including Huang et al. (2025) and Ismayilov et al. (2025), further enhanced StyleGAN2 through attention mechanisms and latent-space manipulation. Similarly, diffusion models have achieved remarkable success in face generation. Denoising Diffusion Probabilistic Models (Ho et al., 2020) and more recent approaches such as Diffusion-driven GAN Inversion (Kim et al., 2024), MMFace-DiT (Krishnamurthy & Rattani, 2026), and MDiTFace (2026) have demonstrated exceptional image quality and diversity. However, both StyleGAN and diffusion-based architectures require substantial computational resources, large-scale datasets, and prolonged training or inference times, limiting their accessibility in resource-constrained environments (Almuammar et al., 2026; Taha, 2026).

Recent surveys further emphasize these challenges. Toshpulatov et al. (2023) identified lightweight attention-augmented DCGAN frameworks as an underexplored research direction, while Cobbinah et al. (2025) confirmed that combining architectural enhancements with regularization consistently yields superior performance. Almuammar et al. (2026) additionally highlighted ongoing challenges related to computational efficiency, scalability, and deployment complexity in modern generative AI systems.

Consequently, there remains a need for computationally efficient generative architectures capable of improving facial realism, structural consistency, feature representation, and adversarial training stability while remaining accessible to resource-constrained research environments. Motivated by these research gaps, this thesis proposes an AGRDCGAN that integrates hierarchical convolutional upsampling, LMHCSA, spectral normalization, and L2 regularization within a unified adversarial learning framework for high-fidelity human face generation.

2.2 Generative Adversarial Networks

The general idea for visual content synthesis is that the general rule is to learn the underlying probability distribution of the data, and then use it to synthesize new and realistic samples. Of the different generative methods, Generative Adversarial Networks have shown to be effective in the generation of photorealistic images.

Generative Adversarial Networks follow an adversarial learning paradigm as a two-player minimax game. In this approach, there are two neural networks which are trained simultaneously: one called the generator and the other called the discriminator . The generator is trained to generate images by mapping random noise vectors from a random latent distribution, whereas the discriminator is trained to determine whether an image is real or fake. Both networks gradually evolve from this competitive process; giving rise to more and more realistic outcomes.

The objective of the training is:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z)))] \quad (2.1)$$

These correspond to the real distribution $p_{\text{data}}(x)$ and the prior distribution over the latent space $p_z(z)$, which is often assumed to be Gaussian or uniform noise. The discriminator $D(x)$ is an estimate of whether a sample is real and the generator $G(z)$ generates synthetic (noise) samples from noise input z. This formulation makes the discriminator learn to make better classification and the generator learn to generate samples that are hard to tell apart from actual data.

Generating human faces is one of the most difficult problems in generative modeling because of the need to capture global facial structure and local facial details like the eyes, nose, mouth and texture of the skin. Moreover, face synthesis should not only guarantee a structural coherence but also guarantee diversity of identity, expression, age and pose.

Even though GAN face generation models are effective, they are subject to problems like mode collapse in which the generator generates limited samples or few faces that decrease the diversity of the face distribution generated. This restriction has a negative impact on realism and generalisation. Despite this, GANs have many applications including virtual avatars, data augmentation, digital entertainment, and film production (Karras et al., 2019).

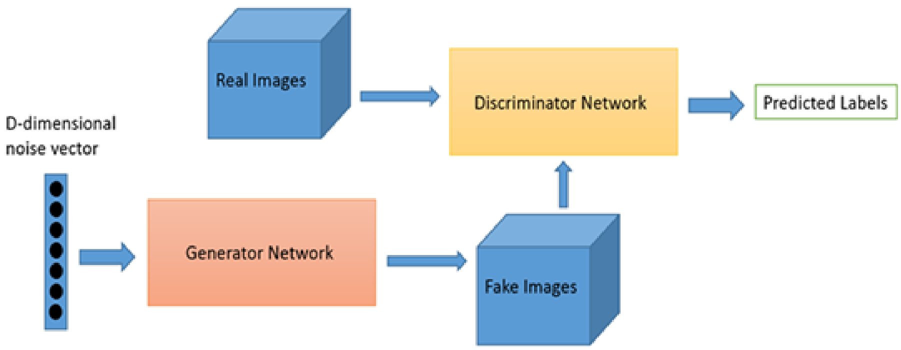

Figure 2.1 GAN Architecture

2.3 Deep Convolutional Generative Adversarial Networks

A major advancement on the architecture of the GAN is the Deep Convolutional Generative Adversarial Networks introduce convolutional neural networks into both generator and discriminator. This integration improves the spatial hierarchy learning capacity of the model in the visual data (Radford et al., 2016). Main improvements to the architecture are based on strided convolutions (downsampling) and transposed convolutions (upsampling).

Because of their generative strength and comparatively easy design, DCGANs are extensively used in generating faces. They are useful for acquiring key facial information like eyes, nose and differences in skin texture (Nekamiche et al., 2022; Lv et al., 2022). The typical DCGAN architectures, however, have a drawback of using local receptive fields that do not have the ability to capture the long range dependencies that are necessary to model the global facial structure and symmetry (Salimans et al., 2016).

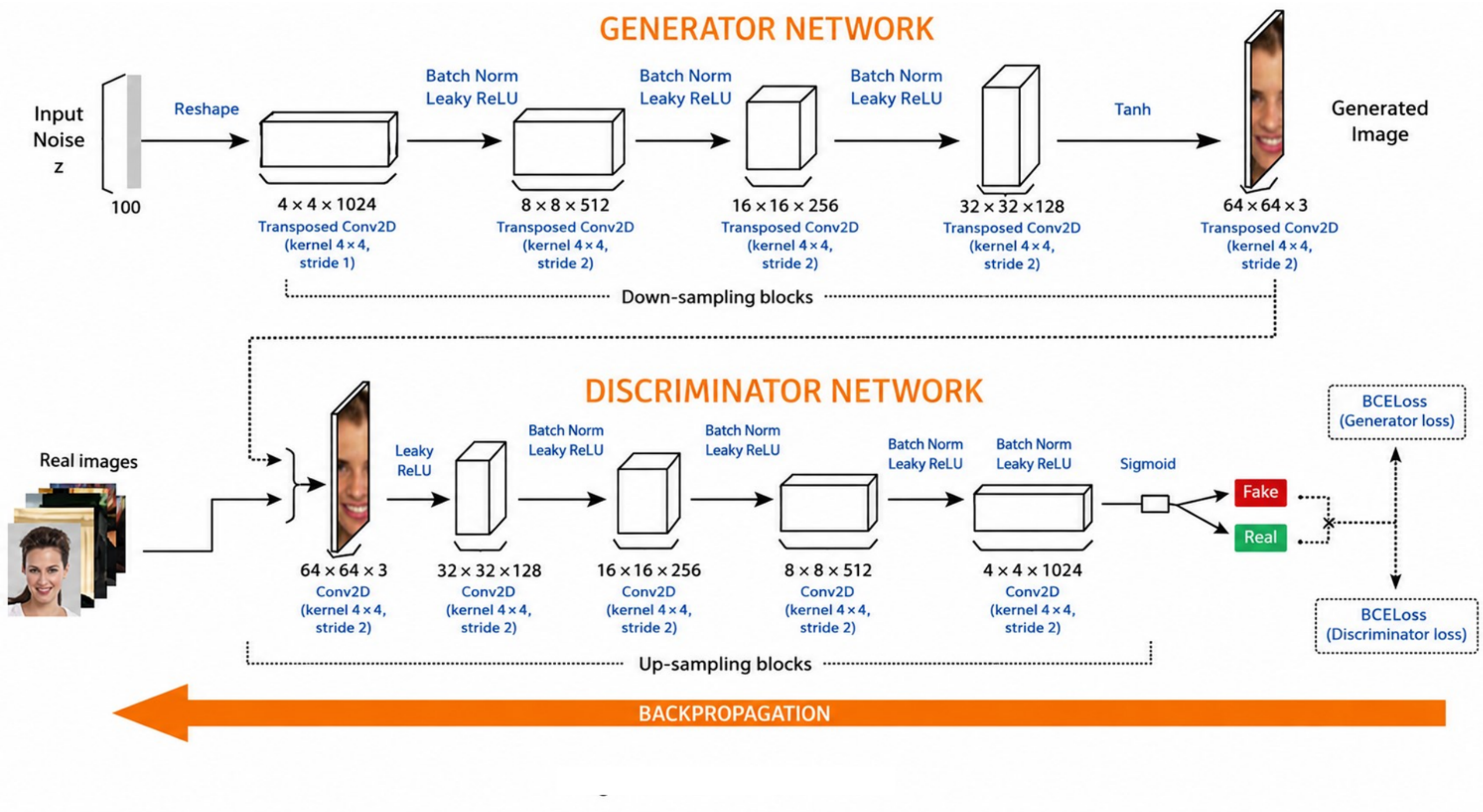

Figure 2.2 DCGAN Architecture (adapted from Nekamiche et al., 2022).

2.4 Attention Mechanisms in Image Generation

2.4.1 Fundamentals of Attention

Attention mechanisms are motivated by the human visual system, which selectively focuses on important regions while suppressing less relevant information. Attention modules are used in deep learning to dynamically assign weights to features or spatial regions, which is useful for better representation of features (Vaswani et al., 2017).

In image generation tasks, it has been demonstrated that attention mechanisms can enhance the consistency and structural alignment across the entire image. Attention can be used to model long-range dependencies between different regions of the face, for instance, between the eyes, nose and mouth, which can be used to improve the symmetry and realism of the animation. (Zhang et al., 2019)

2.4.2 Local Multi-Head Channel Self-Attention

Local Multi-Head Channel Self-Attention is a mechanism that aims to capture the local feature region channel-wise dependencies. Conventional global self-attention models perform attention computation on all spatial positions, while LMHCSA only uses local computation which maintains spatial structure but reduces computational complexity. More than one head is simultaneously trained to attend multiple different sets of feature interactions, across channels (Pecoraro et al., 2022).

LMHCSA dynamically weights channels according to their importance and boosts informative ones while minimizing less informative ones. This is especially useful when generating images of faces, as the texture and structure details drastically affect the perceptual quality. With its localization and lightweight design, LMHCSA can be used in conjunction with convolutional and adversarial architecture to synthesize high fidelity faces (Pecoraro et al., 2022).

2.5 Training Stabilization Techniques

2.5.1 L2 Regularization (Weight Decay)

A common way to enhance generalization and prevent overfitting in deep neural networks is to add what is known as L2 regularization, or weight decay. It does so by introducing a penalty term based on the square of the magnitude of the model weights, to the loss function. This prevents the parameters from being too large, and helps limit the complexity of the model while training.

The optimization task is now a compromise between the prediction error, and the growth of the weights, that reduces the generalization performance measurement (Krogh & Hertz, 1992). L2 regularization helps prevent the model from being too ”noisy” or sensitive to changes in its inputs or noise, by penalizing large weights.

L2 regularization is also crucial in deep networks like convolutional neural networks and generative adversarial networks for stabilizing training. It can help to curb the overpowering nature of the discriminator, thereby impacting adversely on the adversarial learning dynamics (Goodfellow et al., 2016).

L2 regularization encourages smaller weight values and smoother solutions, which is often associated with convergence toward flatter minima that tend to generalize better than sharp minima (Hochreiter & Schmidhuber, 1997; Keskar et al., 2017). Therefore, it is frequently used in conjunction with other stabilisation techniques such as batch normalization and spectral normalization (Srivastava et al., 2014; Ioffe & Szegedy, 2015; Miyato et al., 2018).

The L2 penalty term is given by:

$$L_{L2} = \lambda \sum_i w_i^2 \tag{2.2}$$

Where: L_{L2} is the defined L2 regularization loss term, w_i represents the weight parameter for the i-th model weight, λ is the regularization coefficient (hyperparameter) that controls the strength of the L2 penalty, and $\sum$ denotes the sum over all weight parameters of the model. L2 regularization plays an important role in generative adversarial networks(Mescheder et al., 2018), because it helps to prevent instability during training, such as mode collapse and gradient explosion.

2.5.2 Spectral Normalization

To stabilise the training of generative adversarial network , the Lipschitz constant of the discriminator was introduced in the training process, which was called spectral normalization. It regularises the spectral norms (highest singular values) of every weight matrix, which makes optimization easier and flow of the gradients more stable (Miyato et al., 2018).

In contrast to weight clipping and gradient penalty techniques, spectral normalization is simple to implement and normalises weight matrices directly while training. This enables architectures that can be deep convolutional generative adversarial networks.

On the benchmark datasets, CIFAR-10, STL-10 and ImageNet (Miyato et al., 2018), spectrally normalised GANs have been shown to converge more smoothly, with higher image quality.

One of the benefits of spectral normalization is that it does not let the discriminator be too powerful, which makes it balance out the adversarial training. This produces more stable updates to the generators and better synthesis quality.

Due to these benefits, spectral normalization is commonly used in the current image generation models based on generative adversarial networks, especially when they are used to create high-resolution images with higher training instability.

2.6 Evaluation Metrics for Face Generation

2.6.1 Fréchet Inception Distance

Fréchet Inception Distance is a popular measure of generative models, in particular generative adversarial networks (Heusel et al., 2017). Compares the real and generated image distributions based on the features extracted from a pre-trained Inception network.

The Fréchet Inception Distance score is calculated as:

$$\text{FID} = \|\boldsymbol{\mu}_r - \boldsymbol{\mu}_f\|_2^2 + \text{Tr}\left(\boldsymbol{\Sigma}_r + \boldsymbol{\Sigma}_f - 2(\boldsymbol{\Sigma}_r\boldsymbol{\Sigma}_f)^{1/2}\right) \tag{2.3}$$

Lower FID values indicate closer alignment between the real and generated image distributions, reflecting higher perceptual quality of the generated images.

2.6.2 Complex Wavelet Structural Similarity Index Measure

Complex Wavelet Structural Similarity Index Measure is a perceptual image quality measure to assess the structural similarity between images in the wavelet domain. It is an extension of Structural Similarity Index Measure, using coefficients of complex wavelet, which is more resistant to the geometric distortion (Wang et al., 2004).

The Complex Wavelet Structural Similarity Index Measure is based on comparing the phase of the wavelet coefficients in the image instead of just the difference in pixel intensities. This makes it more in line with the visual perception of humans, particularly in face generation tasks.

It is not sensitive to illumination changes, rotations and small translations, so it is appropriate for use in assessing perceptual similarity for synthetic face images where small variations should not substantially change perceptual similarity.

Complex Wavelet Structural Similarity Index Measure is given below as:

$$\text{CW-SSIM}(\mathbf{x}, \mathbf{y}) = \frac{2\left|\sum_{i=1}^{N} c_{\mathbf{x},i} c_{\mathbf{y},i}^{*}\right| + \delta}{\sum_{i=1}^{N} |c_{\mathbf{x},i}|^2 + \sum_{i=1}^{N} |c_{\mathbf{y},i}|^2 + \delta} \tag{2.4}$$

2.7 Review of Related Work

Generative Adversarial Networks were introduced by Goodfellow et al. (2014), establishing adversarial learning through a generator–discriminator framework capable of modeling complex data distributions. Despite their success, early GANs suffered from training instability, hyperparameter sensitivity, and mode collapse. To improve training stability and image quality, Radford et al. (2016) proposed Deep Convolutional GANs, which introduced convolutional architectures, batch normalization, and stable activation functions. Although DCGAN significantly improved face generation, its convolutional structure limits the modeling of long-range feature dependencies, often resulting in structural inconsistencies and weak facial symmetry. Recent studies such as Nekamiche et al. (2022), Manimegala et al.

(2024), and Lv et al. (2024) demonstrated that integrating lightweight attention and regularization techniques into DCGAN frameworks can improve generation quality and recognition performance, motivating further investigation of efficient GAN architectures.

Several studies have explored techniques for improving image quality and training stability. Progressive image generation strategies were introduced by Karras et al. (2018), demonstrating that hierarchical image synthesis stabilizes adversarial training and supports high-resolution generation. Similarly, spectral normalization (Miyato et al., 2018) and L2 regularization (Krogh & Hertz, 1992) have been widely adopted to improve convergence, reduce overfitting, and stabilize adversarial learning. Kurach et al. (2019) further reported that regularization plays a critical role in determining the effectiveness of attention mechanisms within GAN architectures.

Attention mechanisms have become increasingly important in image generation because of their ability to capture long-range feature dependencies. Self-attention was first introduced for sequence modeling by Vaswani et al. (2017) and later incorporated into image generation through SAGAN by Zhang et al. (2019), improving structural consistency and global feature representation. To reduce computational complexity, Liu et al. (2021) proposed window-based local attention, while Pecoraro et al. (2022) introduced LMHCSA for efficient channel-wise dependency modeling. Li et al. (2023) further demonstrated that attention-guided face generation improves semantic consistency and texture preservation; however, existing approaches rarely combine attention mechanisms with comprehensive stabilization strategies.

StyleGAN-based architectures currently represent the state of the art in high-fidelity face generation. StyleGAN (Karras et al., 2019) and StyleGAN2 (Karras et al., 2020) introduced style-based latent representations that significantly improved image quality, controllability, and identity preservation. Subsequent works such as StyleT2F (Sabae et al., 2022), StyleHEAT (Yin et al., 2022), and attention-enhanced StyleGAN variants (Huang et al., 2025; Ismayilov et al., 2025) further improved realism and controllability. However, these architectures typically require large-scale datasets, extensive computational resources, and long training times.

Diffusion models have emerged as another powerful paradigm for face generation. Denoising Diffusion Probabilistic Models (Ho et al., 2020) demonstrated superior image fidelity and diversity through iterative denoising. Recent studies, including Giambi and Lisanti (2023), Kim et al. (2024), and advanced diffusion transformer models such as MMFace-DiT (Krishnamurthy & Rattani, 2026) and MDiTFace (Cao et al., 2026), achieved impressive face generation performance. Nevertheless, diffusion models generally require hundreds of inference steps, resulting in high computational cost and memory consumption.

Recent survey studies (Toshpulatov et al., 2023; Cobbinah et al., 2025) indicate that while substantial progress has been achieved through GANs, StyleGANs, diffusion models, attention mechanisms, and regularization techniques, lightweight architectures that effectively combine attention and stabilization strategies remain underexplored. Moreover, many state-of-the-art approaches are computationally expensive and difficult to deploy in resource-constrained environments. Motivated by these gaps, this thesis proposes an AGRDCGAN that integrates LMHCSA, spectral normalization, L2 regularization, and progressive convolutional upsampling within a unified and computationally efficient framework for high-fidelity human face generation.

Table 2.1 GAN-Based and DCGAN Face Generation Methods

Paper	**Methodology**	**Key Results**	**Limitations**
Goodfellow et al. (2014)	Vanilla GAN adversarial framework	First adversarial generative model	Unstable training and mode collapse
Radford et al. (2016)	DCGAN with strided convolutions and batch normalization	Stable training and improved image quality	Limited long-range dependency modeling
Nekamiche et al. (2022)	DCGAN-based face generation	Generated recognizable synthetic faces	Weak facial landmark consistency
Manimegala et al. (2024)	DCGAN with SSIM evaluation on CelebA	Improved perceptual quality	Weak global consistency and missing high-frequency details
Lv et al. (2024)	LW-DCGAN with attention modules and L1 regularization	Improved occluded face recognition performance	Limited to occluded face scenarios

Table 2.2 Attention-Based and Style-Based Generation Methods

Paper	**Methodology**	**Key Results**	**Limitations**
Zhang et al. (2019)	Self-Attention GAN (SAGAN)	Improved global structural consistency	High computational cost $O(H^2W^2C)$
Sabae et al. (2022)	StyleT2F using StyleGAN2	Improved semantic text-to-face consistency	Limited adversarial stability
Yin et al. (2022)	StyleHEAT for high-resolution face synthesis	Realistic high-resolution face generation	High computational cost and architectural complexity
Li et al. (2023)	Attention-guided conditional face generation	Improved texture preservation and semantic consistency	Requires conditional inputs and lacks integrated stabilization
Huang et al. (2025)	StyleGAN2 with FreqSplitAttention and Dual Mask Discriminator	Improved facial detail preservation and texture consistency	Specialized for domain-specific face generation tasks

Table 2.3 Regularization, Diffusion, and Proposed Model

Paper	Methodology	Key Results	Limitations
Miyato et al. (2018)	Spectral Normalization	Stable GAN training	Discriminator only
Ho et al. (2020)	DDPM	High realism, strong diversity	Slow inference, computationally expensive
Kim et al. (2024)	Diffusion-GAN Inversion	Multimodal face synthesis	Complex architecture
Krishnamurthy & Rattani (2026)	MMFace-DiT	High-fidelity multimodal faces	Large compute requirement
Cao et al. (2026)	MDiTFace	¿94% computational overhead reduction	Complex diffusion-transformer framework
Proposed AGRDCGAN	DCGAN + LMHCSA + SN + L2	FID=35.73, CW-SSIM=0.4956, Acc.=99.80%	128×128 resolution only

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Overview

This chapter presents the methodological framework of the proposed Attention Guided and regularised Deep Convolutional Generative Adversarial Network with Local Multi-Head Channel Self-Attention for high-fidelity human face generation. The proposed architecture extends the standard Deep Convolutional Generative Adversarial Network (Radford et al., 2016) by integrating a Local Multi-Head Channel Self-Attention mechanism at the 32×32 feature resolution in the generator and 16×16 in the discriminator. In addition, L2 regularization (weight decay) is applied to improve training stability and reduce overfitting and generative artefacts.

The model generated 128×128 RGB face images from 256-dimensional latent noise vectors. The performance of the proposed model is evaluated using Fréchet Inception Distance (Heusel et al., 2017) for measuring perceptual quality and Complex Wavelet Structural Similarity Index Measure (Sampat et al., 2009) for structural similarity assessment.

This chapter is organized as follows: Section 3.2 describes the dataset and preprocessing pipeline; Section 3.3 presents the network architecture including the LMHCSA module; Section 3.4 formulates the loss functions; Section 3.5 explains the regularization techniques; Section 3.6 describes the training procedure; Section 3.7 presents the evaluation metrics; and Section 3.8 discusses downstream evaluation using face verification.

3.2 Dataset and Preprocessing

3.2.1 Dataset Description

The Flickr-Faces-HQ dataset proposed by (Karras et al., 2019) is used as the primary dataset for this thesis. The dataset contains 70,000 high-quality human face images with a resolution of 1024×1024 pixels. FFHQ includes substantial variations in age, gender, ethnicity, facial expression, head pose, illumination conditions, accessories, and background diversity, making it highly suitable for generative adversarial network based face synthesis tasks.

For computational efficiency, a subset of 10,000 images was randomly selected from the full dataset using a fixed random seed of 42 to ensure experimental reproducibility. All selected images were resized using bilinear interpolation to a resolution of 128×128 pixels to preserve facial features while reducing memory requirements.

Figure 3.1 shows some examples of images contained in the FFHQ dataset used this thesis.

Figure 3.1 Sample images of faces from the FFHQ dataset with variations on age, gender, Ethnicity, facial expression, illumination and background conditions.

3.2.2 Data Splitting Strategy

The dataset is split into three subsets: training set, validation set, and test set. Let $N_{\text{total}} = 10{,}000$ denote the total number of samples. The training set is 75% (7,500 images), the validation set is 15% (1,500 images), and the test set is 10% (1,000 images). Splitting has been done with a fixed random seed of 42 for reproducibility. The validation set is utilized for hyperparameter tuning and to track the training performance.

3.2.3 Preprocessing Pipeline

Suppose that an input image is represented by $I_{\text{raw}} \in \mathbb{R}^{H \times W \times 3}$, where H and W are the height and width of the image, respectively. Each image is resized into a 128×128 image by bilinear interpolation, meaning that the value of each output pixel is calculated as a weighted average of its four nearest neighboring input image pixels.

After resizing, pixel values are normalised to the range $[-1, 1]$ using the transformation:

$$I_{\text{norm}} = \frac{I_{\text{resized}}}{127.5} - 1$$

where $I_{\text{resized}} \in [0, 255]$ and $I_{\text{norm}} \in [-1, 1]$. This normalization will be used to make this a compatible layer with Tanh activation function in the generator output layer.

In training, the data augmentation is done on the training set only, with a $p = 0.5$ random horizontal flipping, which is defined as:

$$r \sim \text{Bernoulli}(0.5)$$

If $r = 1$, the image is horizontally flipped; otherwise, it remains unchanged. The validation and test sets are only subjected to resizing and normalization without augmentation.

3.3 Network Architecture

3.3.1 Generator Architecture

The generator $G : \mathbb{R}^Z \rightarrow \mathbb{R}^{128 \times 128 \times 3}$ maps a 256-dimensional latent vector $\mathbf{z}$ to a 128×128 RGB face image. The architecture is based on the Deep Convolutional Generative Adversarial Network (DCGAN) design principles proposed by Radford et al. (2016), and consists of six transposed convolutional blocks with a Local Multi-Head Channel Self-Attention module added after the 32x32 feature resolution block.

The latent vector $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is first passed through a fully connected layer, batch normalization, and ReLU activation functions to generate a 4×4×1024 feature map.A 4×4×1024 feature map is created by first passing the latent vector $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ through a fully connected layer, followed by batch normalization and ReLU activation. This is the initial representation, which is represented as $\mathbf{h}_0 \in \mathbb{R}^{4\times4\times1024}$.

Five transposed convolutional blocks are then used to progressively increase spatial resolution while reducing channel depth. Each block i is defined as:

$$\mathbf{h}_i = \text{ReLU}\left(\text{BN}\left(\text{ConvTranspose2d}_{k=4,s=2,p=1}(\mathbf{h}_{i-1})\right)\right)$$

where k denotes kernel size, s denotes stride, and p denotes padding. The spatial resolution increases as follows: $4\times4\times1024 \rightarrow 8\times8\times512 \rightarrow 16\times16\times256 \rightarrow 32\times32\times128 \rightarrow 64\times64\times64 \rightarrow 128\times128\times32$.

At the 32×32 resolution, the LMHCSA module is applied to capture long-range spatial dependencies and improve structural consistency. After this module, the final upsampling block produces the 128×128 feature map.

The output image is generated using a 3×3 convolution followed by a Tanh activation function:

$$\mathbf{x}_{\text{fake}} = \tanh\left(\text{Conv2d}_{k=3,p=1}(\mathbf{h}_5)\right)$$

where $\mathbf{h}_5 \in \mathbb{R}^{128\times128\times32}$ and $\mathbf{x}_{\text{fake}} \in [-1, 1]^{128\times128\times3}$.

3.3.2 Discriminator Architecture

The discriminator $D : \mathbb{R}^{128\times128\times3} \rightarrow \mathbb{R}$ maps an input image to a scalar value representing its authenticity.

It consists of five convolutional layers that progressively reduce spatial resolution while increasing feature depth. Each convolutional block j is defined as:

$$\mathbf{d}_j = \text{LeakyReLU}_{0.2}\left(\text{BN}\left(\text{Conv2d}_{k=4,s=2,p=1}(\mathbf{d}_{j-1})\right)\right)$$

where LeakyReLU has a negative slope of 0.2. The first convolutional layer does not use batch normalization. The channel progression is $3 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 1024$, while spatial dimensions reduce from 128×128 to 4×4.

At the 16×16 resolution, the LMHCSA module is applied to improve sensitivity to structural inconsistencies in facial regions. All convolutional layers are regularised with spectral normalization (Miyato et al., 2018), helping to stabilize the training process and bound the Lipschitz constant of the discriminator.

The output is finally calculated from a 4×4 convolution:

$$D(\mathbf{x}) = \text{Conv2d}_{k=4,s=1,p=0}(\mathbf{d}_5)$$

When flattened, which yields a scalar logit.

3.3.3 Local Multi-Head Channel Self-Attention

The innovation of this thesis is the Local Multi-Head Channel Self-Attention module. LMHCSA considers global self-attention, but uses attention only locally to reduce the computational cost while maintaining spatial structure. It is derived from the Swin Transformer (Liu et al., 2021), which has a window-based attention mechanism.

Given an input feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, where B denotes batch size, C denotes number of channels, and H, W denote spatial dimensions, the LMHCSA module operates as follows.

Step 1: QKV Projection. Query, Key, and Value tensors are computed using 1×1 convolutions:

$$\mathbf{Q} = \mathbf{W}_q \circledast \mathbf{X}, \quad \mathbf{K} = \mathbf{W}_k \circledast \mathbf{X}, \quad \mathbf{V} = \mathbf{W}_v \circledast \mathbf{X}$$

Step 2: Multi-Head Splitting. The channel dimension is divided into $h = 4$ heads, each with dimension $d_h = C/h$.

Step 3: Attention Computation. Scaled dot-product attention is computed as:

$$\text{Attention} = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}}\right)\mathbf{V}_h$$

Step 4: Output Projection. Outputs from all heads are concatenated and projected:

$$\mathbf{O} = \mathbf{W}_o \circledast \text{Concat}(\text{head}_1, \ldots, \text{head}_h)$$

Step 5: Residual Connection. The final output is:

$$\mathbf{X}_{out} = \mathbf{X} + \gamma \cdot \mathbf{O}$$

where γ is a learnable scaling parameter initialized to zero.

The computational complexity of LMHCSA is significantly lower than global self-attention, reducing from $O(H^2W^2C)$ to approximately $O(HWC)$, making it more efficient for high-resolution face generation tasks.

3.4 Loss Function

The proposed model employs the standard non-saturating adversarial loss as introduced in the original Generative Adversarial Network framework (Goodfellow et al., 2014), implemented via binary cross-entropy with logits.

3.4.1 Discriminator Loss

Let $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ denote a real image sampled from the empirical data distribution, and $\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})$ denote a 256-dimensional latent vector sampled from a standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. The discriminator aims to maximise the probability of correctly classifying real and generated images. The discriminator loss is defined as:

$$\mathcal{L}_D = -\mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}[\log D(\mathbf{x})] - \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}}[\log(1 - D(G(\mathbf{z})))]$$

where $\mathbb{E}[\cdot]$ denotes expectation, $D(\mathbf{x}) \in (0, 1)$ is the discriminator's estimated probability that $\mathbf{x}$ is real (obtained using a sigmoid activation), and $G(\mathbf{z})$ is the generated image.

3.4.2 Discriminator Loss with Label Smoothing

To improve training stability, label smoothing(Szegedy et al., 2016) is applied. Let $y_{\text{real}} = 0.9$ and $y_{\text{fake}} = 0.1$ be the smoothed labels for real and fake samples, respectively. The discriminator loss is defined as:

$$\mathcal{L}_D = -\frac{1}{m}\sum_{i=1}^{m}\Big[y_{\text{real}}\log(\sigma(D(\mathbf{x}_i))) + (1 - y_{\text{real}})\log(1 - \sigma(D(\mathbf{x}_i)))\Big]$$
$$-\frac{1}{m}\sum_{i=1}^{m}\Big[y_{\text{fake}}\log(\sigma(D(G(\mathbf{z}_i)))) + (1 - y_{\text{fake}})\log(1 - \sigma(D(G(\mathbf{z}_i))))\Big]$$

where $\mathcal{L}_D$ is the discriminator loss function, m is the batch size representing the number of samples in a mini-batch, $\mathbf{x}_i$ is a real sample from the dataset, $\mathbf{z}_i$ is a random latent vector sampled from the distribution $p_{\mathbf{z}}$, $G(\cdot)$ is the generator function that maps a latent vector to a synthetic image, $D(\cdot)$ is the discriminator function that outputs a real-valued score, $\sigma(\cdot)$ is the sigmoid activation function defined as $\sigma(x) = \frac{1}{1+e^{-x}}$, $y_{\text{real}} = 0.9$ is the smoothed label for real images, and $y_{\text{fake}} = 0.1$ is the smoothed label for fake images.

3.4.3 Generator Loss

The generator aims to fool the discriminator by generating images that are classified as real. The generator loss is defined as:

$$\mathcal{L}_G = -\mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}}[\log\sigma(D(G(\mathbf{z})))]$$

In practice, this is implemented using binary cross-entropy over a mini-batch:

$$\mathcal{L}_G = -\frac{1}{m}\sum_{i=1}^{m}\log\sigma(D(G(\mathbf{z}_i)))$$

where $\mathcal{L}_G$ is the generator loss function, $\mathbf{z}_i$ is the latent noise vector sampled from the distribution $p_{\mathbf{z}}$, $G(\mathbf{z}_i)$ is the generated image from the generator, $D(\cdot)$ is the discriminator output, and m is the batch size.

3.5 Regularisation Techniques

3.5.1 Spectral Normalisation

Spectral Normalisation is applied to every convolutional layer in the discriminator of the proposed AGRDCGAN. It was introduced by Miyato et al. (2018) to stabilise adversarial training.

For a weight matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, spectral normalisation is defined as:

$$\mathbf{W}_{\mathrm{SN}} = \frac{\mathbf{W}}{\sigma(\mathbf{W})}$$

where $\mathbf{W}$ is the weight matrix of a convolutional layer in the discriminator, $\mathbf{W}_{\mathrm{SN}}$ is the spectrally normalized weight matrix, and $\sigma(\mathbf{W})$ is the largest singular value (spectral norm) of $\mathbf{W}$.

The spectral norm $\sigma(\mathbf{W})$ is computed using the power iteration method.

This normalization enforces a Lipschitz constraint on the discriminator:

$$\|D(\mathbf{x}_1) - D(\mathbf{x}_2)\|_2 \leq \|\mathbf{x}_1 - \mathbf{x}_2\|_2$$

where $D(\cdot)$ is the discriminator function, $\mathbf{x}_1, \mathbf{x}_2$ are the two input samples, and $\|\cdot\|_2$ denotes the Euclidean (L2) norm.

By constraining the Lipschitz constant, spectral normalisation stabilises gradient flow, reduces gradient explosion, and improves convergence during adversarial training.

3.5.2 L2 Regularisation

L2 regularisation (also known as weight decay) is applied to the generator of the proposed Attention-guided and regularised Deep Convolutional Generative Adversarial Network to penalize large weight magnitudes.

The regularisation term is defined as:

$$\mathcal{L}_{\mathrm{reg}} = \lambda\|\mathbf{W}\|_2^2 = \lambda \sum_{i=1}^{m} \sum_{j=1}^{n} W_{ij}^2$$

where $\mathcal{L}_{\mathrm{reg}}$ is the L2 regularization loss, $\lambda = 1 \times 10^{-4}$ is the regularization coefficient, $\mathbf{W}$ is the set of trainable weight parameters of the generator, and W_{ij} denotes an individual weight parameter at position (i, j).

The total generator loss is:

$$\mathcal{L}_G^{\mathrm{total}} = \mathcal{L}_G + \lambda\|\mathbf{W}_G\|_2^2$$

where $\mathcal{L}_G^{\mathrm{total}}$ is the total generator loss, $\mathcal{L}_G$ is the adversarial generator loss, and $\mathbf{W}_G$ represents the generator weight parameters.

L2 regularisation reduces overfitting, improves generalization, and helps prevent artefacts such as chequerboard patterns and high-frequency noise in generated images. The discriminator does not use L2 regularisation ($\lambda_D = 0$) to maintain strong discriminative capability.

3.5.3 Batch Normalisation

Batch normalisation (Ioffe & Szegedy, 2015) is applied after the transposed convolutional layers in the generator and after convolutional layers 2–5 in the discriminator.

For a mini-batch of m samples, let $\mu_{\mathcal{B}}$ denote the batch mean and $\sigma_{\mathcal{B}}^2$ denote the batch variance. The normalization process is defined as:

$$\hat{x}_i = \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}}, \qquad y_i = \gamma \hat{x}_i + \beta$$

where x_i is the input activation of the i^{th} sample, $\hat{x}_i$ is the normalized activation, $\mu_{\mathcal{B}}$ is the mean of the mini-batch, $\sigma_{\mathcal{B}}^2$ is the variance of the mini-batch, $\epsilon = 1 \times 10^{-5}$ is a small constant for numerical stability, γ is the learnable scale parameter, β is the learnable shift parameter, and y_i is the output after batch normalization.

Batch normalization reduces internal covariate shift, accelerates convergence, and improves training stability in deep generative adversarial networks.

3.6 Training Procedure

3.6.1 Hyperparameter Configuration

The following hyperparameters are used throughout all experiments:

where the batch size is $m = 16$ images per iteration, the total number of epochs is $E = 85$, the generator learning rate is $\eta_G = 2 \times 10^{-4}$, the discriminator learning rate is $\eta_D = 8 \times 10^{-5}$, the Adam optimizer is utilized with hyper-parameters $\beta_1 = 0.5, \beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-8}$, the generator weight decay is $\lambda_G = 1 \times 10^{-4}$, the discriminator weight decay is $\lambda_D = 0$, the latent dimension is $Z = 256$, and the number of discriminator update steps per generator step is $n_{\text{critic}} = 1$. The use of a higher generator learning rate compared to the discriminator helps maintain a balanced adversarial learning process, which improves training stability as discussed in Heusel et al. (2017).

3.6.2 Training Algorithm

For each epoch $e = 1, 2, \ldots, E$, the model iterates over all mini-batches of the training dataset. For each batch of real images $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m\}$, the following steps are performed:

Discriminator Update: A batch of latent vectors $\{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_m\}$ is sampled from a standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. The generator produces fake samples $\mathbf{x}_{\text{fake}}^{(i)} = G(\mathbf{z}_i)$. The discriminator computes predictions for both real and fake images, and the discriminator loss $\mathcal{L}_D$ is computed using binary cross-entropy with label smoothing (0.9 for real samples and 0.1 for fake samples). The discriminator parameters are then updated using one step of the Adam optimizer.

Generator Update: A new batch of latent vectors is sampled, and corresponding fake images are generated. The generator loss $\mathcal{L}_G$ is computed using binary cross-entropy where the target label is 1, encouraging the generator to produce images that the discriminator classifies as real. The generator parameters are updated using one step of the Adam optimizer.

No additional loss functions such as gradient penalty, feature matching, or perceptual loss are used in this baseline configuration, enabling a direct evaluation of the proposed Attention guided DCGAN architecture.

3.6.3 Validation and Model Selection

The model was evaluated every 5 epochs using the validation dataset. During evaluation, the generator produces images from fixed latent vectors to ensure consistent comparison across epochs. Two evaluation metrics are used:

- **FID:** Measures the distance between real and generated image feature distributions. Lower values indicate better image quality (Heusel et al., 2017).
- **CW-SSIM:** Evaluates structural similarity in the wavelet domain. Higher values indicate better structural consistency (Sampat et al., 2009).

The generator model achieving the lowest validation FID is selected as the best model and saved as `best_enhanced_dcgan.pth`. This checkpoint is used for final evaluation on the test dataset.

3.7 Evaluation Metrics

3.7.1 Fréchet Inception Distance

Heusel et al. (2017) proposed the Fréchet Inception Distance to measure the distance between feature distributions of real and generated images. A pre-trained Inception v3 network (Szegedy et al., 2016) is used as a feature extractor, where the final classification layer is removed, producing a 2,048-dimensional feature representation.

Let $\mathcal{N}(\boldsymbol{\mu}_r, \boldsymbol{\Sigma}_r)$ and $\mathcal{N}(\boldsymbol{\mu}_f, \boldsymbol{\Sigma}_f)$ denote Gaussian distributions fitted to real and generated feature embeddings, respectively. The FID is defined as:

$$\text{FID} = \|\boldsymbol{\mu}_r - \boldsymbol{\mu}_f\|_2^2 + \text{Tr}\left(\boldsymbol{\Sigma}_r + \boldsymbol{\Sigma}_f - 2(\boldsymbol{\Sigma}_r\boldsymbol{\Sigma}_f)^{1/2}\right)$$

Where: $\boldsymbol{\mu}_r$ is the mean feature vector of real images, $\boldsymbol{\mu}_f$ is the mean feature vector of generated (fake) images, $\boldsymbol{\Sigma}_r$ is the covariance matrix of real image features, $\boldsymbol{\Sigma}_f$ is the covariance matrix of generated image features, $\|\boldsymbol{\mu}_r - \boldsymbol{\mu}_f\|_2^2$ denotes the squared Euclidean distance between the mean vectors, $\text{Tr}(\cdot)$ represents the trace operator (sum of diagonal elements), and $(\boldsymbol{\Sigma}_r\boldsymbol{\Sigma}_f)^{1/2}$ is the matrix square root of the product of the covariance matrices.

Lower FID values indicate higher similarity between real and generated distributions.In this thesis, FID is computed using 1500 real and 1500 generated samples from the validation set.

3.7.2 Complex Wavelet Structural Similarity Index

Sampat et al. (2009) introduced the Complex Wavelet Structural Similarity Index to evaluate structural similarity in the complex wavelet domain. Using four decomposition scales and the `db4` wavelet basis, CW-SSIM is robust to small translations and rotations, making it suitable for face image evaluation.

Let $c_{\mathbf{x},i}$ and $c_{\mathbf{y},i}$ denote complex wavelet coefficients of images $\mathbf{x}$ and $\mathbf{y}$, where $i = 1, 2, \ldots, N$. CW-SSIM is defined as:

$$\text{CW-SSIM}(\mathbf{x}, \mathbf{y}) = \frac{2\left|\sum_{i=1}^{N} c_{\mathbf{x},i} c_{\mathbf{y},i}^{*}\right| + \delta}{\sum_{i=1}^{N} |c_{\mathbf{x},i}|^2 + \sum_{i=1}^{N} |c_{\mathbf{y},i}|^2 + \delta}$$

Where: $c_{\mathbf{x},i}$ is the complex wavelet coefficient of image $\mathbf{x}$ at position i, $c_{\mathbf{y},i}$ is the complex wavelet coefficient of image $\mathbf{y}$ at position i, $c_{\mathbf{y},i}^{*}$ is the complex conjugate of $c_{\mathbf{y},i}$, $|\cdot|$ denotes the complex modulus (magnitude), N is the total number of wavelet coefficients, and $\delta = 10^{-6}$ is a small constant for numerical stability.

CW-SSIM scores range from 0 to 1. Higher scores mean structural similarity.

3.8 Downstream Task: Face Verification

To assess the practical utility of the generated face images, a downstream face verification experiment was conducted. Face verification is a binary classification task that determines whether two face images belong to the same identity. This evaluation provides an objective measure of the degree to which the proposed AGRDCGAN model preserves identity discriminative facial features in the generated images.

3.8.1 Task Definition

Let $\mathbf{x}_a, \mathbf{x}_b \in \mathbb{R}^{128\times128\times3}$ be two normalised face images. A similarity function is learned:

$$s : (\mathbf{x}_a, \mathbf{x}_b) \to [0, 1]$$

Where: $\mathbf{x}_a, \mathbf{x}_b$ are the input face images, s is the similarity score between 0 and 1, and $y \in \{0, 1\}$ is the ground truth label where 1 indicates the same identity and 0 indicates a different identity.

3.8.2 Datasets

Three datasets are used for comparison:

- $\mathcal{D}_{\text{real}}$: 1,000 real LFW test images - $\mathcal{D}_{\text{enhanced}}$: 1,000 images generated by the proposed model - $\mathcal{D}_{\text{vanilla}}$: 1,000 images generated by baseline DCGAN

3.8.3 Experimental Protocol

The evaluation follows five steps:

Step 1 – Pair Construction: From each dataset, 250 positive and 250 negative pairs are sampled.

Step 2 – Feature Learning: A ResNet18 model pretrained on ImageNet is fine-tuned using contrastive loss.

Step 3 – Distance Computation: Euclidean distance between embeddings is computed:

$$d_i = \|F(\mathbf{x}_a^{(i)}) - F(\mathbf{x}_b^{(i)})\|_2$$

Where: d_i is the Euclidean distance between embeddings for pair i, $F(\cdot)$ is the embedding function mapping an image to a feature vector, $\mathbf{x}_a^{(i)}, \mathbf{x}_b^{(i)}$ represents the i-th image pair, and $\|\cdot\|_2$ denotes the Euclidean (L2) norm.

Step 4 – Threshold Selection: Optimal threshold τ^* is obtained using Youden's J statistic:

$$\tau^* = \arg\max_{\tau} (\mathrm{TPR}(\tau) - \mathrm{FPR}(\tau))$$

Where: τ^* is the optimal decision threshold, $\mathrm{TPR}(\tau)$ is the True Positive Rate at threshold τ, $\mathrm{FPR}(\tau)$ is the False Positive Rate at threshold τ, and $\arg\max$ denotes the argument that maximizes the expression.

Step 5 – Evaluation: Final performance is reported using accuracy and ROC-AUC:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}_i = y^{(i)}], \quad \text{AUC} = \int_0^1 \mathrm{TPR}(t)\, d\mathrm{FPR}(t)$$

Where: N is the total number of test pairs, $\hat{y}_i$ is the predicted label for pair i, $y^{(i)}$ is the ground truth label for pair i, $\mathbf{1}[\cdot]$ is the indicator function which returns 1 if the condition is true and 0 otherwise, $\mathrm{TPR}(t)$ is the True Positive Rate at threshold t, $\mathrm{FPR}(t)$ is the False Positive Rate at threshold t, and AUC is the Area Under the ROC Curve.

CHAPTER FOUR

PROPOSED ARCHITECTURE

4.1 Overview of Proposed Architecture

This chapter presents the complete architectural design of the proposed Attention-guided and regularised Deep Convolutional Generative Adversarial Network with Local Multi-Head Channel Self-Attention. The proposed architecture extends the standard Deep Convolutional Generative Adversarial Network framework (Radford et al., 2016) through three major modifications: (1) integration of Local Multi-Head Channel Self-Attention modules at resolution-specific locations in both the generator and discriminator, (2) application of L2 regularization (weight decay) to the generator to improve training stability and reduce generative artefacts, and (3) spectral normalization applied to all convolutional layers in the discriminator. The complete model is designed to generate 128×128 RGB face images from 256-dimensional latent noise vectors sampled from a standard normal distribution.

Figure 4.1 presents a high-level overview of the proposed architecture. The framework consists of two adversarial networks: a generator (G) that synthesizes realistic face images from random noise vectors, and a discriminator (D) that distinguishes real images from generated images. The generator progressively upsamples a 256-dimensional latent vector into a 128×128 RGB face image using transposed convolutional layers. To improve feature dependency modeling, the LMHCSA module is integrated at the 32×32 feature resolution. The discriminator progressively downsamples input images using strided convolutional layers to produce a single classification logit, while LMHCSA attention is integrated at the 16×16 feature resolution to enhance structural feature extraction. Spectral normalization is additionally applied to all discriminator convolutional layers to stabilise adversarial training.

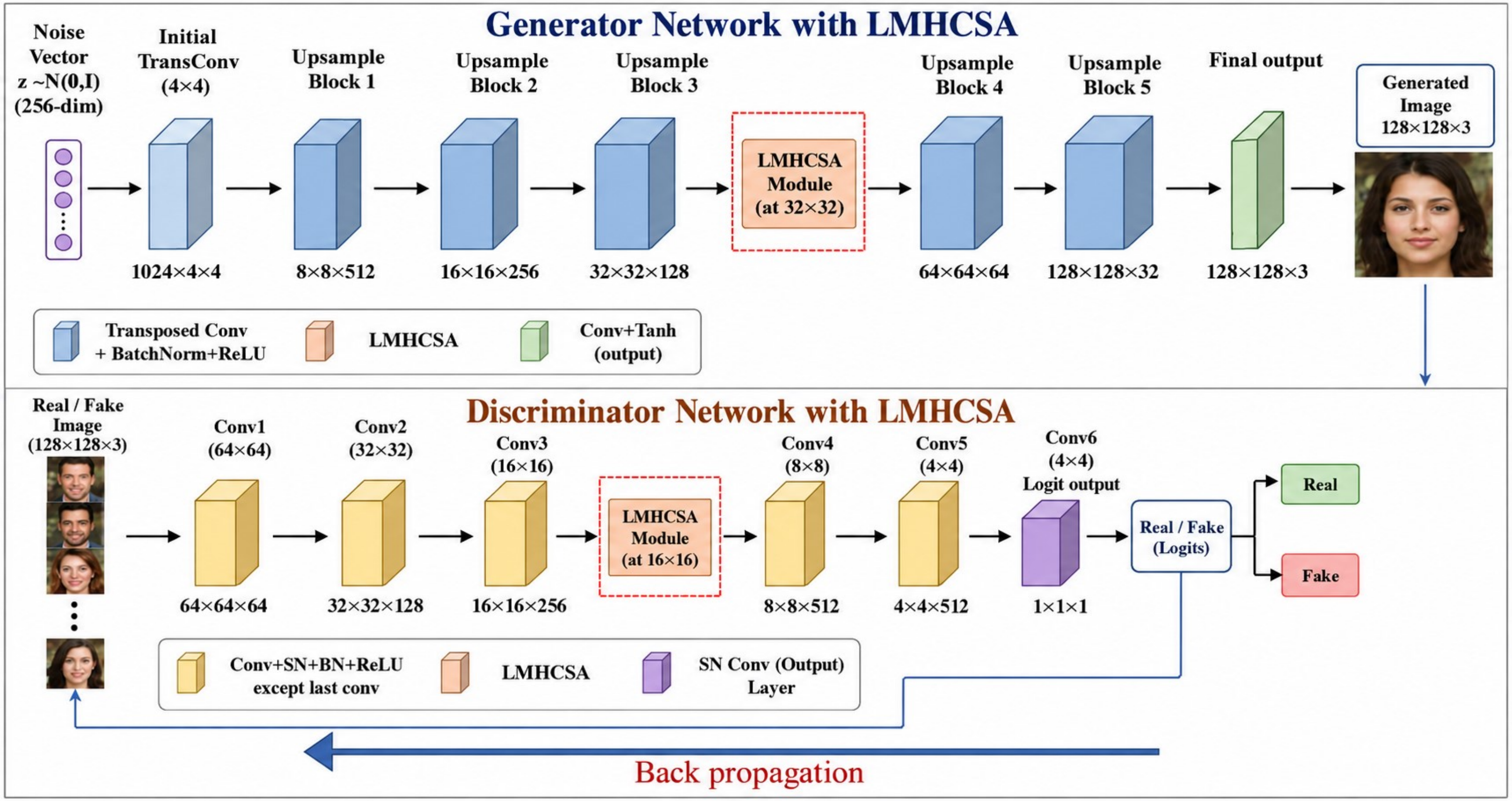

Figure 4.1 Proposed architecture of the Attention-guided and regularised DCGAN

Table 4.1 High-level configuration of the proposed Attention-guided and regularised DCGAN

Parameter	**Value**
Input latent dimension (Z)	256
Output image resolution	$128 \times 128 \times 3$
Generator attention resolution	32×32 (128 channels)
Discriminator attention resolution	16×16 (256 channels)
Number of attention heads (h)	4
Generator L2 regularization (λ_G)	1×10^{-4}
Discriminator L2 regularization (λ_D)	0.0
Parameter Counts	
Generator parameters	15,439,139
Discriminator parameters	11,427,585
Total parameters	26,866,724

4.2 Generator Architecture

The generator $G : \mathbb{R}^Z \rightarrow \mathbb{R}^{128\times128\times3}$ maps a latent vector $\mathbf{z} \in \mathbb{R}^{256}$ to a synthetic face image. The generator follows a progressive upsampling strategy consisting of six transposed convolutional blocks and a single LMHCSA attention module inserted at the 32×32 feature resolution.

Figure 4.2 illustrates the complete generator architecture, showing the transformation from the 256-dimensional latent vector to the final 128×128 RGB output image.

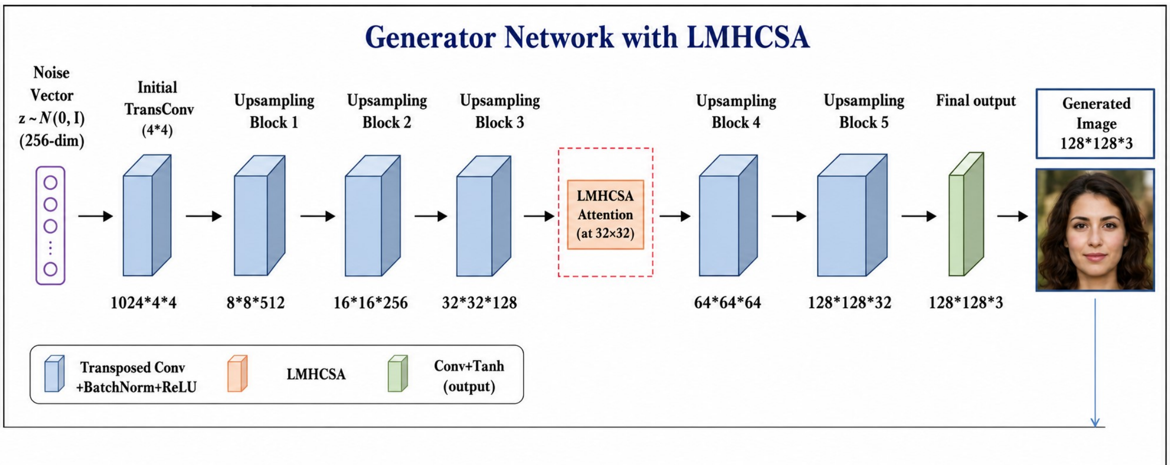

Figure 4.2 Generator architecture of the proposed Attention-guided and regularised DCGAN. The latent vector $\mathbf{z}$ is projected to a $4 \times 4 \times 1024$ feature map and progressively upsampled through transposed convolutional blocks. LMHCSA attention is applied at the 32×32 feature resolution. The final convolution layer with Tanh activation produces the 128×128 RGB output image.

4.2.1 Latent Space Representation

Let $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{256})$ denote a 256-dimensional latent vector sampled from the multivariate standard normal distribution, where $\mathbf{I}_{256} \in \mathbb{R}^{256\times256}$ represents the identity covariance matrix. The latent vector is reshaped into a spatial tensor:

$$\mathbf{z}_{\text{spatial}} = \text{reshape}(\mathbf{z}) \in \mathbb{R}^{B\times256\times1\times1}$$

where B denotes the batch size, 256 represents the latent feature channels, and 1×1 is the initial spatial resolution.

This spatial representation enables subsequent convolutional operations.

4.2.2 Initial Projection Layer

The reshaped latent vector is passed through a transposed convolutional layer to produce a $4 \times 4 \times 1024$ feature map:

$$\mathbf{h}_0 = \text{ReLU}\left(\text{BN}\left(\text{ConvTranspose2d}_{k=4,s=1,p=0}(\mathbf{z}_{\text{spatial}})\right)\right)$$

where ConvTranspose2d denotes the transposed convolution operation, $k = 4$ is the kernel size, $s = 1$ is the stride, $p = 0$ is the padding, BN denotes batch normalization, and $\text{ReLU}(x) = \max(0, x)$ is the Rectified Linear Unit activation function.

The output feature map is:

$$\mathbf{h}_0 \in \mathbb{R}^{B\times1024\times4\times4}$$

which serves as the initial spatial representation for subsequent upsampling stages.

4.2.3 Upsampling Blocks

Five sequential transposed convolutional blocks progressively increase spatial resolution while reducing channel dimensionality. For each block $i \in \{1, 2, 3, 4, 5\}$, the operation is defined as:

$$\mathbf{h}_i = \text{ReLU}\left(\text{BN}\left(\text{ConvTranspose2d}_{k=4,s=2,p=1}(\mathbf{h}_{i-1})\right)\right)$$

where $\mathbf{h}_{i-1}$ is the input feature map from the previous block, $\mathbf{h}_i$ is the output feature map, $k = 4$ is the kernel size, $s = 2$ is the stride for spatial upsampling, and $p = 1$ is the padding size.

The transposed convolution with stride $s = 2$ doubles the spatial resolution at each stage. The complete feature progression is:

$\mathbf{h}_0$	$: 4 \times 4 \times 1024$	Initial projection
$\mathbf{h}_1$	$: 8 \times 8 \times 512$	Upsampling block 1
$\mathbf{h}_2$	$: 16 \times 16 \times 256$	Upsampling block 2
$\mathbf{h}_3$	$: 32 \times 32 \times 128$	Upsampling block 3
$\mathbf{h}_4$	$: 64 \times 64 \times 64$	Upsampling block 4
$\mathbf{h}_5$	$: 128 \times 128 \times 32$	Upsampling block 5

4.2.4 LMHCSA Attention at 32 x 32 Resolution

At the 32×32 resolution, where $\mathbf{h}_3 \in \mathbb{R}^{B\times128\times32\times32}$, the LMHCSA attention module is applied as:

$$\mathbf{h}_3^{\text{attn}} = \text{LMHCSA}(\mathbf{h}_3)$$

The attention module is inserted at this resolution to capture channel-wise feature dependencies while maintaining computational efficiency. At the 32×32 feature scale, important facial structures such as the eyes, nose, and mouth are sufficiently represented for attention-based feature refinement.

The LMHCSA module enhances the generator's ability to model long-range feature relationships and improve structural consistency in generated face images. The complete mathematical formulation of the LMHCSA module is presented in Section 4.4.

4.2.5 Output Layer

Following the attention module, two additional upsampling blocks generate the final feature map $\mathbf{h}_5 \in \mathbb{R}^{B\times32\times128\times128}$. A final convolution layer with Tanh activation produces the RGB output image:

$$\mathbf{x}_{\text{fake}} = \tanh\left(\text{Conv2d}_{k=3,p=1}(\mathbf{h}_5)\right)$$

where $\text{Conv2d}_{k=3,p=1}$ denotes a convolution layer with kernel size $k = 3$ and padding $p = 1$, and $\tanh(\cdot)$ constrains the output values to the range $[-1, 1]$.

The output $\mathbf{x}_{\text{fake}} \in \mathbb{R}^{B\times3\times128\times128}$ represents a batch of generated RGB face images.

4.3 Discriminator Architecture

The discriminator $D : \mathbb{R}^{128\times128\times3} \to \mathbb{R}$ is designed as a binary classifier that distinguishes real facial images from synthetic images generated by the proposed generator network. The discriminator follows a progressive downsampling strategy consisting of five convolutional blocks and a single LMHCSA attention module inserted at the 16×16 feature resolution.

Figure 4.3 illustrates the complete discriminator architecture, showing the transformation from the 128×128 RGB input image to the final real/fake prediction score.

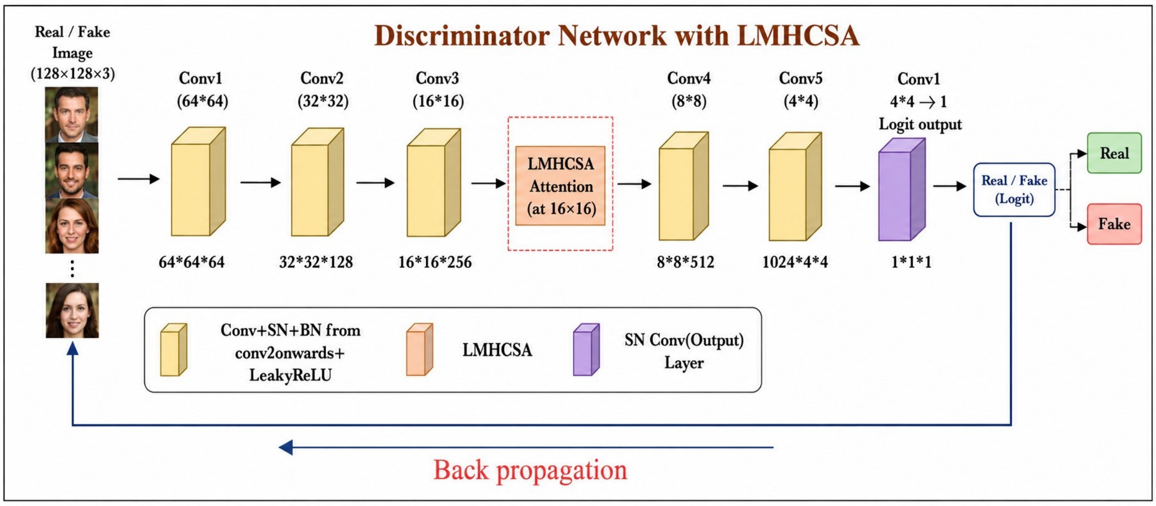

Figure 4.3 Discriminator architecture of the proposed Attention-guided and regularised DC-GAN. The input image is progressively downsampled through convolutional blocks with spectral normalization. LMHCSA attention is applied at the 16×16 feature resolution. The final convolution layer produces a single scalar logit representing the probability that the input image is real or fake.

4.3.1 Input Representation

The discriminator receives an RGB face image represented as:

$$\mathbf{x} \in \mathbb{R}^{B \times 3 \times 128 \times 128}$$

where: where B denotes the batch size, 3 represents the RGB image channels, and 128×128 is the spatial image resolution.

The input image is progressively transformed into compact hierarchical feature representations through convolutional operations.

4.3.2 Downsampling Convolutional Blocks

The discriminator employs five sequential convolutional blocks to progressively reduce spatial dimensions while increasing feature depth. Each block consists of spectral-normalised convolution, batch normalization, and LeakyReLU activation. The operation of each block is defined as:

$$\mathbf{d}_i = \text{LeakyReLU}\left(\text{BN}\left(\text{SN}\left(\text{Conv2d}_{k=4,s=2,p=1}(\mathbf{d}_{i-1})\right)\right)\right)$$

where $\mathbf{d}_{i-1}$ is the input feature map, $\mathbf{d}_i$ is the output feature map, SN denotes spectral normalization, BN denotes batch normalization, $k = 4$ is the kernel size, $s = 2$ is the stride, and $p = 1$ is the padding size.

The first convolutional block does not use batch normalization, following the standard

DCGAN discriminator design.

The complete feature progression through the discriminator is:

$$\begin{aligned} \mathbf{d}_0 &: 128 \times 128 \times 3 && \text{Input image} \\ \mathbf{d}_1 &: 64 \times 64 \times 64 && \text{Conv block 1} \\ \mathbf{d}_2 &: 32 \times 32 \times 128 && \text{Conv block 2} \\ \mathbf{d}_3 &: 16 \times 16 \times 256 && \text{Conv block 3} \\ \mathbf{d}_4 &: 8 \times 8 \times 512 && \text{Conv block 4} \\ \mathbf{d}_5 &: 4 \times 4 \times 1024 && \text{Conv block 5} \end{aligned}$$

Each convolutional block doubles the number of feature channels while reducing spatial resolution by a factor of two. This hierarchical structure enables the discriminator to learn both low-level textures and high-level semantic facial structures.

4.3.3 LMHCSA Attention at 16 x 16 Resolution

At the 16×16 feature resolution, where

$$\mathbf{d}_3 \in \mathbb{R}^{B \times 256 \times 16 \times 16},$$

the LMHCSA attention module is applied as:

$$\mathbf{d}_3^{\text{attn}} = \text{LMHCSA}(\mathbf{d}_3)$$

The LMHCSA module is inserted at this intermediate feature scale to capture long-range spatial dependencies and channel-wise feature relationships while maintaining computational efficiency.

At the 16×16 resolution, important global facial structures such as eye alignment, facial symmetry, nose shape, and mouth positioning are effectively represented. The attention mechanism improves the discriminator's ability to detect structural inconsistencies and synthetic artefacts in generated images.

The complete mathematical formulation of the LMHCSA module is presented in Section 4.4.

4.3.4 Output Layer

Following the attention module and the final convolutional blocks, a 4×4 convolution layer with stride 1 and zero padding produces the final discriminator output:

$$D(\mathbf{x}) = \text{Conv2d}_{k=4,s=1,p=0}(\mathbf{d}_5)$$

The output tensor has shape:

$$(B, 1, 1, 1)$$

which is reshaped into:

$$(B, 1)$$

to produce a scalar logit for each input image.

The discriminator does not apply a sigmoid activation function because the binary cross-entropy with logits loss (`BCEWithLogitsLoss`) internally combines sigmoid activation and loss computation for improved numerical stability during adversarial training.

Positive logits indicate high confidence that the input image is real, while negative logits indicate that the image is synthetic.

4.3.5 Spectral Normalisation

For every convolutional layer in the discriminator, spectral normalisation (Miyato et al., 2018) is applied. For a weight matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, spectral normalisation replaces $\mathbf{W}$ with:

$$\mathbf{W}_{\mathrm{SN}} = \frac{\mathbf{W}}{\sigma(\mathbf{W})}$$

where $\sigma(\mathbf{W})$ denotes the largest singular value of $\mathbf{W}$, computed efficiently via power iteration. This normalisation enforces a Lipschitz constraint:

$$\|D(\mathbf{x}_1) - D(\mathbf{x}_2)\|_2 \leq \|\mathbf{x}_1 - \mathbf{x}_2\|_2$$

for any two inputs $\mathbf{x}_1, \mathbf{x}_2$. By bounding the Lipschitz constant to 1, spectral normalisation stabilises adversarial training and prevents gradient explosion.

4.4 Local Multi-Head Channel Self-Attention

LMHCSA module is the core architectural contribution of this thesis. Unlike global self-attention, which computes pairwise relationships across all spatial positions and incurs quadratic computational complexity $O(H^2W^2C)$, the proposed LMHCSA mechanism applies attention using multi-head channel representations while preserving spatial locality. Our design helps reduce computing costs and improves how we model feature dependencies. The formulation is inspired by multi-head attention introduced by Vaswani et al. (2017).

4.4.1 Input Feature Representation

Let the input feature map be represented as:

$$\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W}$$

where B denotes the batch size, C denotes the number of feature channels, H denotes the height of the feature map, and W denotes the width of the feature map.

In the proposed architecture, the LMHCSA module operates on: where 32×32 feature maps with $C = 128$ channels in the generator, and 16×16 feature maps with $C = 256$ channels in the discriminator.

The objective of the LMHCSA module is to compute attention-weighted feature representations while preserving the original spatial structure through residual learning.

4.4.2 Query, Key, and Value Projection

Three independent 1×1 convolution layers are applied to generate the Query, Key, and Value tensors:

$$\mathbf{Q} = \mathbf{W}_q \circledast \mathbf{X}, \qquad \mathbf{K} = \mathbf{W}_k \circledast \mathbf{X}, \qquad \mathbf{V} = \mathbf{W}_v \circledast \mathbf{X}$$

where $\circledast$ denotes the convolution operation, $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{C \times C \times 1 \times 1}$ are learnable convolution kernels, and $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{B \times C \times H \times W}$.

The use of 1×1 convolutions enables channel transformation without altering spatial dimensions.

4.4.3 Multi-Head Channel Decomposition

The channel dimension is divided into h attention heads. The feature dimension of each head is defined as:

$$d_h = \frac{C}{h}$$

where h denotes the number of attention heads, and d_h denotes the feature dimension of each head.

In this thesis, $h = 4$.

The Query, Key, and Value tensors are reshaped as:

$$\begin{aligned}
\mathbf{Q}_h &= \text{reshape}(\mathbf{Q}) \in \mathbb{R}^{B \times h \times d_h \times N} \\
\mathbf{K}_h &= \text{reshape}(\mathbf{K}) \in \mathbb{R}^{B \times h \times d_h \times N} \\
\mathbf{V}_h &= \text{reshape}(\mathbf{V}) \in \mathbb{R}^{B \times h \times d_h \times N}
\end{aligned}$$

where $N = H \times W$ denotes the total number of spatial positions, $\mathbf{Q}_h$ denotes the reshaped Query tensor, $\mathbf{K}_h$ denotes the reshaped Key tensor, and $\mathbf{V}_h$ denotes the reshaped Value tensor.

The tensors are subsequently permuted to the format:

$$(B, h, N, d_h)$$

to facilitate efficient batch matrix multiplication during attention computation.

4.4.4 Attention Computation

For each attention head, scaled dot-product attention is computed as:

$$\text{Attention}(\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h) = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}}\right)\mathbf{V}_h$$

where $\mathbf{Q}_h \mathbf{K}_h^\top \in \mathbb{R}^{B\times h\times N\times N}$ denotes the attention score matrix, $\mathbf{K}_h^\top$ denotes the transpose of the Key tensor, $\sqrt{d_h}$ is a scaling factor used to stabilise gradient magnitudes, and softmax$(\cdot)$ converts attention scores into normalised attention weights.

The attention mechanism enables the model to learn contextual relationships between spatial feature locations while maintaining computational efficiency.

4.4.5 Output Projection

The outputs from all attention heads are concatenated along the channel dimension:

$$\mathbf{O}_{\text{concat}} = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)$$

where:

$$\mathbf{O}_{\text{concat}} \in \mathbb{R}^{B\times C\times H\times W}$$

A final 1×1 convolution is then applied:

$$\mathbf{O} = \mathbf{W}_o \circledast \mathbf{O}_{\text{concat}}$$

where $\mathbf{W}_o \in \mathbb{R}^{C\times C\times 1\times 1}$ is a learnable projection matrix, and $\mathbf{O}$ denotes the final attention output.

4.4.6 Residual Feature Fusion

The attention output is integrated with the original input feature map using a residual connection:

$$\mathbf{X}_{\text{out}} = \mathbf{X} + \gamma\mathbf{O}$$

where $\mathbf{X}$ is the input feature map from the previous layer, $\mathbf{O}$ is the output of the attention module representing the refined feature map produced by the LMHCSA mechanism, γ is a learnable scaling parameter that controls the contribution of the attention output, and $\mathbf{X}_{\text{out}}$ is the final output feature map obtained by combining the input features with the scaled attention output through a residual connection.

Initialising $\gamma = 0$ allows the module to begin as an identity mapping, enabling stable optimisation during early training stages. As training progresses, the network gradually learns the contribution of attention-based features.

4.4.7 Computational Complexity Analysis

The computational complexity of conventional global self-attention applied over spatial dimensions is:

$$\mathcal{O}_{\text{global}}(H^2W^2C)$$

For a feature map of size 32×32 with $C = 128$ channels, the complexity becomes approximately:

$$32^4 \times 128 \approx 134 \text{ million operations}$$

In contrast, the proposed LMHCSA module achieves:

$$\mathcal{O}_{\text{LMHCSA}}(HW \cdot h \cdot d_h)$$

For the same feature dimensions ($H = W = 32$, $h = 4$, $d_h = 32$), the computational cost is approximately:

$$32 \times 32 \times 4 \times 32 \approx 131 \text{ thousand operations}$$

This represents approximately a 1000× reduction in computational complexity compared with conventional global self-attention while still enabling effective contextual feature modelling.

CHAPTER FIVE

IMPLEMENTATION DETAILS

5.1 Implementations

This chapter presents the implementation details of the proposed Attention guided and regularised DCGAN model for high-fidelity human face generation. It describes the experimental environment, software configuration, dataset preprocessing, hyperparameter settings, model architecture (generator and discriminator), LMHCSA attention module, training procedure, evaluation metrics, and experimental setup. The complete source code is provided in Appendix A.

5.2 Working Environment

This section describes the hardware configuration used for training and evaluating the proposed model.

Table 5.1 Hardware Tools Specification

Computer Component	**Specification**
Processor	Intel(R) Core(TM) i7-8700M CPU @ 3.20GHz 3.19 GHz
Installed RAM	8.00 GB
Operating System	Windows 10
System Type	64-bit operating system, x64-based processor
Graphics	Intel(R) HD Graphics 520 / NVIDIA Corporation
GPU	NVIDIA GeForce RTX 2070 SUPER
Driver Version	560.94
CUDA Version	12.6
GPU Memory	8192 MB (8 GB)

5.3 Environmental Setup

We used software tools and environments that work with PyTorch to help develop, train and evaluate our model.

We used Anaconda as our environment to manage Python packages and dependencies. This made it easy to install libraries like NumPy, pandas and Matplotlib. We used version 23.7.4 of Anaconda.

Jupyter Notebook gave us a space to combine code, documentation and visuals. This made it easier to repeat experiments and debug.

Kaggle was used to access free cloud-based GPU resources and to perform supplementary experiments as well as dataset handling and comparison with existing approaches.

Finally, a local GPU workstation was used for full-scale training of the proposed model for 85 epochs.

Table 5.2 summarises the key software packages and versions used in this thesis.

Table 5.2 Software Environment Specifications

Package	**Version**
Python	3.10.0
PyTorch	2.0.0+
CUDA Toolkit	12.6
TorchVision	0.15.0
NumPy	1.24.0
Pillow (PIL)	9.0.0
Matplotlib	3.6.0
scikit-learn	1.2.0
SciPy	1.10.0
tqdm	4.65.0
pyiqa	0.1.0+

5.4 Random Seed Configuration

To ensure reproducibility of experimental results, a fixed random seed was used across all stochastic operations. Table 5.3 presents the random seed configuration adopted in this thesis.

Table 5.3 Random Seed Configuration for Reproducibility

Component	**Configuration**
Seed value	42
PyTorch CPU	`torch.manual_seed(42)`
PyTorch CUDA	`torch.cuda.manual_seed(42)`
PyTorch CUDA (all devices)	`torch.cuda.manual_seed_all(42)`
NumPy	`np.random.seed(42)`
Python random module	`random.seed(42)`
CuDNN deterministic mode	`True`
CuDNN benchmark mode	`False`

5.5 Dataset Implementation

We used the Flickr-Faces-HQ dataset, which has 70,000 quality human face images. Each image has a resolution of 1024×1024 pixels.

For efficiency we randomly picked 10,000 images from the dataset. We used a fixed seed value of 42 to ensure we could repeat the process. The dataset was then divided into three mutually exclusive subsets: 75% for training (7,500 images), 15% for validation (1,500 images), and 10% for testing (1,000 images), as summarized in Table 5.4.

Table 5.4 Dataset Statistics After Splitting

Dataset Split	**Number of Images**
Training set (75%)	7,500
Validation set (15%)	1,500
Test set (10%)	1,000
Total	10,000

5.5.1 Preprocessing and Data Augmentation

The preprocessing and data augmentation strategies applied to the dataset are summarized in Table 5.5.

Table 5.5 Preprocessing and Data Augmentation Configuration

Operation	**Training Set**	**Validation/Test Set**
Resize to 128×128	Yes	Yes
Normalization to [-1, 1]	Yes	Yes
Random horizontal flip (p = 0.5)	Yes	No

All images were resized to 128×128 pixels using bilinear interpolation. The pixel values were normalised from the range [0, 255] to [-1, 1] using the transformation:

$$I_{\text{norm}} = \frac{I_{\text{resized}}}{127.5} - 1$$

For the training set, random horizontal flipping was applied with a probability of $p = 0.5$.

5.5.2 Model Training Hyperparameters

Table 5.6 presents the hyperparameter configuration used for training the proposed LMHCSA-DCGAN model.

In addition to the selected configuration, several hyperparameter settings were experimentally evaluated, including latent dimensions (128, 256), batch sizes (8, 16,), and learning

Table 5.6 Model Training Hyperparameters

Parameter	Value
Image size (IMG_SIZE)	128 × 128 pixels
Latent dimension (Z_DIM)	256
Batch size (BATCH_SIZE)	16
Number of epochs (EPOCHS)	85
Generator learning rate (LR_G)	0.0002
Discriminator learning rate (LR_D)	0.00008
Generator weight decay	0.0001
Discriminator weight decay	0.0
Adam optimizer β_1	0.5
Adam optimizer β_2	0.999
Loss function	Binary Cross-Entropy with Logits

rates. These configurations were tested to analyze model stability and performance variations. The final hyperparameters were selected based on training stability and the quality of generated face images.

5.6 Implementation of LMHCSA Attention Module

The Local Multi-Head Channel Self-Attention module is a part of our thesis. It does attention along the channel dimension, which helps keep locality. This design reduces computational costs while enabling effective modeling of inter-channel feature dependencies.

5.6.1 LMHCSA Configuration

Table 5.7 presents the configuration parameters of the LMHCSA module.

Table 5.7 LMHCSA Attention Module Configuration

Parameter	Value
Number of attention heads (h)	4
Generator channel dimension	128
Discriminator channel dimension	256
Learnable residual scaling parameter (γ)	Initialized to 0
QKV projection kernel size	1 × 1
Output projection kernel size	1 × 1
Attention scaling factor	$\sqrt{d_h}^{-1}$

The LMHCSA module works by:

The module operates by computing Query, Key, and Value feature maps through 1×1 convolutions, splitting them into multiple attention heads, applying scaled dot-product attention across heads, and aggregating the outputs through a residual connection.

We present the mathematical formulation of LMHCSA, in Chapter 4. The complete implementation is provided in Appendix.A.3.

5.6.2 Generator Architecture

The generator network $G : \mathbb{R}^{256} \rightarrow \mathbb{R}^{128\times128\times3}$ is designed to map a 256-dimensional latent vector $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{256})$ into a high-resolution RGB facial image. The model follows a deep convolutional upsampling strategy using transposed convolution layers.

The input latent vector is first projected into a high-dimensional feature representation and reshaped into a 4×4 spatial feature map. This representation is progressively upsampled through a series of convolutional blocks to construct the final image. To improve global feature dependency and spatial coherence, a Local Multi-Head Channel Self-Attention module is introduced at the 32×32 resolution stage.

Table 5.8 summarizes the generator architecture and parameter distribution.

Table 5.8 Generator Architecture (Transposed Convolution Blocks)

Layer Type	**Details**	**Output Size**
Input Projection	ConvTranspose2d(256 → 1024, kernel=4, stride=1, padding=0) + BatchNorm + ReLU	4×4
Hidden Block 1	ConvTranspose2d(1024 → 512, kernel=4, stride=2, padding=1) + BatchNorm + ReLU	8×8
Hidden Block 2	ConvTranspose2d(512 → 256, kernel=4, stride=2, padding=1) + BatchNorm + ReLU	16×16
Hidden Block 3	ConvTranspose2d(256 → 128, kernel=4, stride=2, padding=1) + BatchNorm + ReLU	32×32
Attention Module	LMHCSA (128 channels, 4 heads)	32×32
Hidden Block 4	ConvTranspose2d(128 → 64, kernel=4, stride=2, padding=1) + BatchNorm + ReLU	64×64
Hidden Block 5	ConvTranspose2d(64 → 32, kernel=4, stride=2, padding=1) + BatchNorm + ReLU	128×128
Output Layer	Conv2d(32 → 3, kernel=3, padding=1) + Tanh	128×128

The final output layer uses a Tanh activation function to normalise pixel values into the range $[-1, 1]$. All layers in the generator are fully trainable. The LMHCSA attention module enhances long-range spatial dependencies, improves structural consistency, and contributes to generating more realistic facial details.

5.7 Implementation of Discriminator Architecture

The discriminator network $D : \mathbb{R}^{128\times128\times3} \to \mathbb{R}$ maps an input image to a single scalar logit, where higher values indicate real images and lower values indicate generated (fake) images.

The architecture follows a deep convolutional downsampling design with spectral normalization applied to all convolutional layers to stabilise adversarial training. An LMHCSA attention module is integrated at the 16×16 resolution to enhance global feature dependency.

Table 5.9 summarizes the discriminator architecture.

Table 5.9 Discriminator Architecture with Parameter Distribution

Layer	**Configuration**	**Output Size**
Conv Block 1	Conv2d(3 $\to$ 64, k=4, s=2, p=1) + SpectralNorm + LeakyReLU	64×64
Conv Block 2	Conv2d(64 $\to$ 128, k=4, s=2, p=1) + BatchNorm + SpectralNorm + LeakyReLU	32×32
Conv Block 3	Conv2d(128 $\to$ 256, k=4, s=2, p=1) + BatchNorm + SpectralNorm + LeakyReLU	16×16
Attention Module	LMHCSA (256 channels, 4 heads)	16×16
Conv Block 4	Conv2d(256 $\to$ 512, k=4, s=2, p=1) + BatchNorm + SpectralNorm + LeakyReLU	8×8
Conv Block 5	Conv2d(512 $\to$ 1024, k=4, s=2, p=1) + BatchNorm + SpectralNorm + LeakyReLU	4×4
Output Layer	Conv2d(1024 $\to$ 1, k=4, s=1, p=0) + SpectralNorm	1×1

5.8 Model Parameter Summary

Table 5.10 presents the total number of trainable parameters in each network component. All parameters in both the generator and discriminator are trainable, as no layers were frozen during training.

Table 5.10 Model Parameter Summary

Model	Component	Parameters
Generator	Initial projection	4,196,352
	Block 1	8,389,632
	Block 2	2,097,664
	Block 3	524,544
	LMHCSA attention	66,048
	Block 4	131,200
	Block 5	32,832
	Output layer	867
Generator Total		**15,439,139**
Discriminator	Conv1	3,072
	Conv2	131,328
	Conv3	524,800
	LMHCSA attention	263,169
	Conv4	2,098,176
	Conv5	8,390,656
	Final convolution	16,384
Discriminator Total		**11,427,585**
Combined Total		**26,866,724**

The parameter distribution shows that the majority of parameters in the generator are concentrated in the early transposed convolution blocks, particularly Block 1 and Block 2, which account for the largest share of the model capacity. In the discriminator, most parameters are concentrated in the deeper convolutional layers, especially Conv4 and Conv5, which capture high-level semantic features.

The LMHCSA attention modules contribute a relatively small portion of the total parameters in both networks, indicating that the proposed attention mechanism is parameter-efficient while still enhancing feature representation and global dependency modeling.

5.9 Training Procedure

The training process follows the standard adversarial learning framework of Generative Adversarial Networks. In this proposed model, the discriminator is trained to correctly distinguish between real and generated images, while the generator is trained to produce images that can fool the discriminator.

5.9.1 Loss Function

Binary Cross-Entropy (BCE) with logits is used as the objective function for both the generator and discriminator. BCE measures the difference between predicted probabilities and true binary labels, penalizing incorrect predictions using a logarithmic loss function that improves training stability.

The Binary Cross-Entropy (BCE) loss for a single sample is defined as:

$$\mathrm{BCE}(y, p) = -\left[y \log(p) + (1 - y) \log(1 - p)\right]$$

where y is the ground-truth label and p is the predicted probability.

Using this definition, the discriminator is trained to distinguish between real and generated samples. Its loss function is defined as:

$$\begin{aligned} L_D = &-\frac{1}{m}\sum_{i=1}^{m}\left[y_{\text{real}} \log \sigma(D(x^{(i)})) + (1 - y_{\text{real}}) \log(1 - \sigma(D(x^{(i)})))\right] \\ &-\frac{1}{m}\sum_{i=1}^{m}\left[y_{\text{fake}} \log \sigma(D(G(z^{(i)}))) + (1 - y_{\text{fake}}) \log(1 - \sigma(D(G(z^{(i)}))))\right] \end{aligned} \tag{5.1}$$

where, L_D denotes the discriminator loss, m denotes the batch size or number of samples in a mini-batch, i denotes the index of each sample in the batch, $x^{(i)}$ denotes the i^{th} real input image sample, $z^{(i)}$ denotes the i^{th} random noise vector sampled from the latent space, $G(z^{(i)})$ denotes the generated (fake) image produced by the generator from the input noise vector, $D(\cdot)$ denotes the discriminator function that estimates the probability that an input sample is real, $\sigma(\cdot)$ denotes the sigmoid activation function used to map the discriminator output into a probability value between 0 and 1, y_{real} denotes the smoothed label assigned to real samples (e.g., 0.9 instead of 1.0), y_{fake} denotes the label assigned to generated samples (e.g., 0.0), and $\log(\cdot)$ denotes the natural logarithm function.

The generator aims to fool the discriminator by making generated samples appear real. Its loss function is defined as:

$$L_G = -\frac{1}{m}\sum_{i=1}^{m} \log \sigma(D(G(z^{(i)}))) \tag{5.2}$$

$$L_G^{\text{total}} = L_G + \lambda \|W_G\|_2^2 \tag{5.3}$$

where, L_G denotes the generator loss, m denotes the batch size or number of samples in a mini-batch, i denotes the index of each sample in the batch, $z^{(i)}$ denotes the i^{th} random noise vector sampled from the latent space, $G(z^{(i)})$ denotes the generated image produced by the generator from the input noise vector, $D(\cdot)$ denotes the discriminator function that estimates the probability that an input sample is real, $\sigma(\cdot)$ denotes the sigmoid activation function that maps the discriminator output into a probability value between 0 and 1, L_G^{total} denotes the total generator loss after regularization, λ denotes the regularization coefficient that controls the strength of the L2 penalty, W_G denotes the trainable weight parameters of the generator network, and $\|W_G\|_2^2$ denotes the squared L2 norm of the generator weights used for regularization.

5.9.2 Training Configuration

Table 5.11 summarizes the optimizer settings used for both networks.

Table 5.11 Optimizer Configuration

Parameter	**Generator**	**Discriminator**
Optimizer	Adam	Adam
Learning rate	0.0002	0.00008
β_1	0.5	0.5
β_2	0.999	0.999
Weight decay (L2)	0.0001	0.0

5.9.3 Validation Strategy

The model is evaluated every 5 epochs using the validation dataset. The evaluation metrics include the Fréchet Inception Distance and the Complex Wavelet Structural Similarity, which measure the distributional similarity and structural fidelity of generated images, respectively. The model with the lowest validation FID is selected as the best-performing checkpoint.

5.10 Implementation of Evaluation Metrics

5.10.1 Fréchet Inception Distance

The Fréchet Inception Distance (Heusel et al., 2017) measures the distance between the feature distributions of real and generated images. A pretrained Inception v3 network (Szegedy et al., 2016) is used as a feature extractor by removing its final classification layer, producing a 2048-dimensional feature representation.

To evaluate the generator we use the FID score. We compute it using 1,500 images and 1,500 generated images. This way we ensure that the evaluation is statistically reliable. The complete PyTorch implementation for initializing the modified Inception v3 network and calculating the evaluation metric is provided in Appendix 7.7.

5.10.2 Complex Wavelet Structural Similarity Index

The Complex Wavelet Structural Similarity Index (Sampat et al., 2009) evaluates structural similarity in the complex wavelet domain. It uses a multi-scale decomposition (four levels) with the Daubechies-4 (`db4`) wavelet basis.

This metric is robust to small geometric distortions such as translations and rotations, making it particularly suitable for evaluating generated facial images. The implementation is based on the `pyiqa` library, with the metric initialization and batch evaluation loop structured as shown in Appendix 7.8.

5.11 Computational Resources

Table 5.12 summarizes the computational resources utilized during model training.

Table 5.12 Computational Resource Usage

Resource	Usage
GPU memory usage	Approximately 6–7 GB
Training time (85 epochs)	Approximately 11 hours and 13 minutes
Dataset storage (subset)	$\approx$ 5 GB
Model checkpoints storage	$\approx$ 500 MB
Generated samples storage	$\approx$ 100 MB

5.12 Experimental Setup

Table 5.13 presents the experimental configurations used to evaluate the proposed method.

Table 5.13 Experimental Configurations

SN	Experimental Setting	Dataset Used
1	Baseline DCGAN for face generation	FFHQ subset (10,000 images)
2	DCGAN with LMHCSA attention module	FFHQ subset (10,000 images)
3	Proposed LMHCSA-DCGAN (full model)	FFHQ subset (10,000 images)

The experimental configurations in Table 5.13 are designed to evaluate the contribution of each component of the proposed model. In the first experiment, a standard DCGAN is trained as a baseline model to provide a performance reference.

In the second experiment, the LMHCSA attention module is integrated into the DCGAN architecture at the 32×32 resolution in the generator and at the 16×16 resolution in the discriminator to enhance feature representation. Finally, the third experiment combines both the LMHCSA attention mechanism and L2 regularization, forming the complete proposed LMHCSA-DCGAN model.

CHAPTER SIX

RESULT AND DISCUSSION

6.1 Chapter Overview

This chapter presents the experimental results of the proposed AGRDCGAN model for high-fidelity human face generation. The performance of the model is evaluated using both quantitative metrics (FID and CW-SSIM) and qualitative visual analysis. A comparative study is conducted between the proposed AGRDCGAN model and the baseline Vanilla DCGAN. Furthermore, a downstream evaluation is performed using a face verification task on the LFW dataset to assess the practical effectiveness of the generated images for data augmentation in biometric recognition systems. The findings are analysed and discussed in detail at the end of the chapter.

6.2 Experimental Setup

All experiments were conducted using the hardware and software environment described in Chapter 5. Both the proposed AGRDCGAN and the baseline Vanilla DCGAN were trained under identical experimental conditions to ensure a fair comparison. The training configuration is summarised in Table 6.1.

Table 6.1 Experimental Conditions for Model Evaluation

Parameter	**Value**
Training epochs	85
Batch size	16
Learning rate (Generator)	0.0002
Learning rate (Discriminator)	0.00008
Validation frequency	Every 5 epochs
Evaluation metrics	FID, CW-SSIM
Downstream task	Face verification (Accuracy, ROC-AUC)

6.3 Training Dynamics Analysis

This section analyses the training behaviour of both the proposed AGRDCGAN and the baseline Vanilla DCGAN over 85 epochs. Understanding training dynamics is essential for evaluating convergence stability and adversarial balance between generator and discriminator networks.

6.3.1 Generator Loss Progression and Adversarial Stability

The generator loss serves as a proxy for the adversarial "tension" within the training process, representing the network's ability to synthesize samples that are indistinguishable from the ground-truth distribution. Figures 6.1 and 6.2 provide a comparative visualization of this progression for the AGRDCGAN and Vanilla DCGAN architectures.

Analysis of Proposed AGRDCGAN Dynamics

The training dynamics of the AGRDCGAN model reveal a well-balanced learning process. Training initiates with a generator loss of 8.62, which rapidly declines to approximately 4.0 within the first 20 epochs. At this stage, the generator successfully captures global facial geometry, such as overall face shape and initial feature placement. Between epochs 20 and 80, the generator loss experiences an upward trend, peaking at 6.02. In adversarial training, a rising generator loss typically indicates that the discriminator is becoming more proficient. However, the integration of Spectral Normalization prevents the discriminator from over-powering the generator too quickly, thereby forcing the generator to continuously refine its outputs and produce highly detailed images. During the final epochs (81,85), the loss slightly decreases and stabilizes at 5.77. This stabilization signifies that the LMHCSA attention mechanism has matured, enabling the model to generate intricate fine-grained textures and structures. Ultimately, the synergy between the generator, Spectral Normalization, and the LMHCSA mechanism allows the generator to effectively deceive the discriminator in the final stages of convergence.

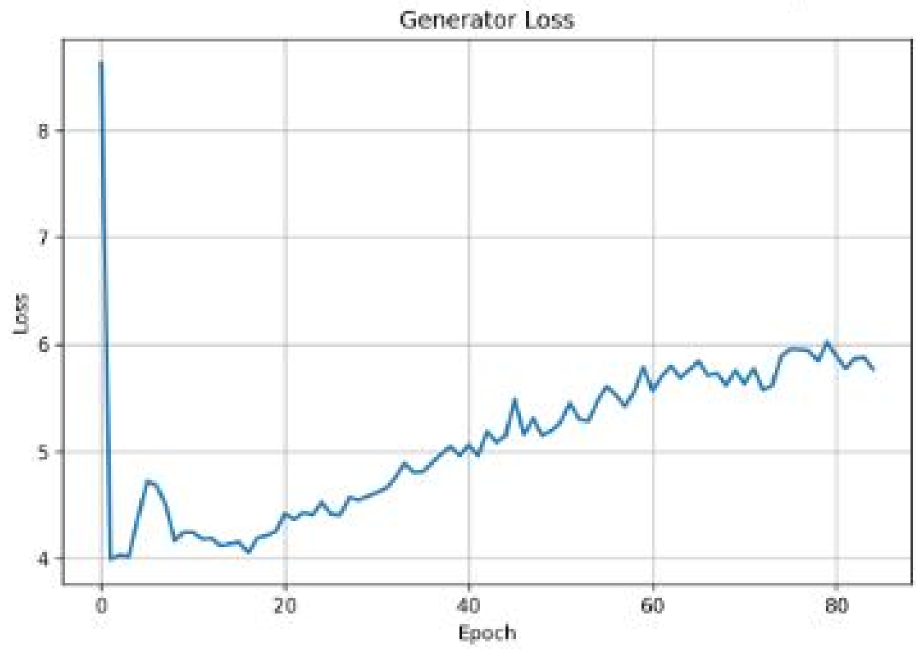

Figure 6.1 Generator training loss progression for AGRDCGAN over 85 epochs. The curve demonstrates initial feature acquisition followed by an adversarial climb and late-stage refinement at 5.77.

Comparison with Vanilla DCGAN Baseline

Figure 6.2 displays the generator loss for the Vanilla DCGAN baseline. This model demonstrates significantly higher initial volatility, starting with a loss exceeding 11.0 and undergoing an extreme plunge to a local minimum of 2.8 near Epoch 18. This sharp drop often correlates with early-stage mode collapse or an imbalanced learning rate where the generator temporarily dominates a weaker discriminator.

Unlike the proposed model, the Vanilla DCGAN lacks a terminal refinement phase, maintaining a linear climb toward 6.0 at the conclusion of training. This suggests that the baseline architecture struggled to achieve the same degree of adversarial equilibrium as the AGRDCGAN. The high-frequency oscillations throughout the baseline training further confirm a less stable training environment compared to the proposed architecture's more controlled progression.

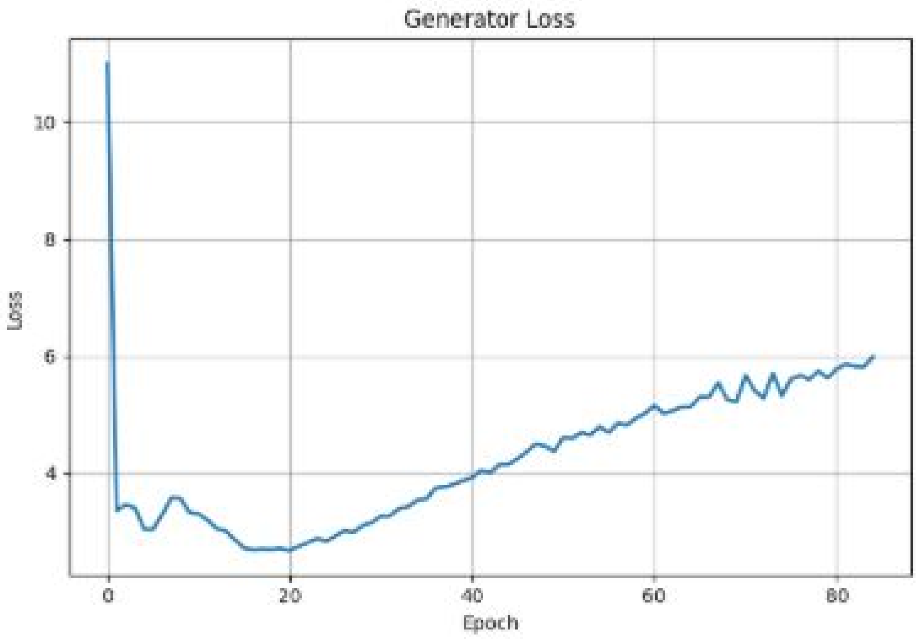

Figure 6.2 Generator training loss progression for Vanilla DCGAN. The extreme initial spike and erratic oscillations indicate lower training stability compared to the AGRDCGAN.

Table 6.2 Generator Loss Comparison: AGRDCGAN vs. Vanilla DCGAN

Metric	AGRDCGAN	Vanilla DCGAN	Observation
Initial Loss (Epoch 1)	**8.62**	11.0	Reduced Initial Volatility
Terminal Loss (Epoch 85)	**5.77**	6.00	Optimized Convergence
Min. Convergence Point	4.00	2.80	Stable Feature Acquisition
Late-Stage Refinement	Yes	No	High-Fidelity Optimization

6.3.2 Discriminator Loss Progression and Adversarial Stability

The discriminator loss serves as a critical indicator of the network's ability to differentiate between synthesized samples and the ground-truth distribution. A stable, gradually decreasing discriminator loss is ideal, as it indicates a healthy adversarial competition where the generator is forced to continuously improve.

Proposed AGRDCGAN Discriminator Analysis

Figure 6.3 presents the discriminator loss curve for the proposed AGRDCGAN model.The loss initiated at 0.91 (Epoch 1) and demonstrated a consistent, controlled descent throughout the 85 epochs of training. By the terminal stage, the loss settled at approximately 0.13

The smoothness of this curve was a direct result of the integration of **Spectral Normalization**, which constrains the Lipschitz constant of the discriminator. This prevents the discriminator from becoming too powerful or "overconfident" early in training, thereby avoiding the vanishing gradient problem. The sustained downward trend without significant spikes indicates a well-balanced adversarial relationship, providing the generator with a reliable and stable learning signal.

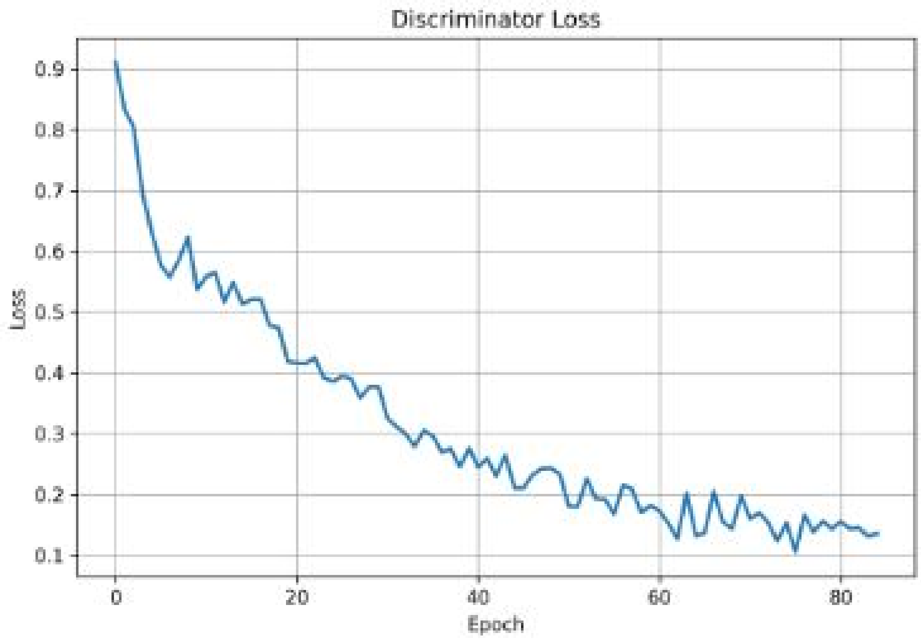

Figure 6.3 Discriminator training loss progression for AGRDCGAN. The model demonstrates a stable descent from 0.91 to 0.13, indicating effective adversarial balancing.

Vanilla DCGAN Baseline Comparison

Figure 6.4 illustrates the discriminator loss for the Vanilla DCGAN. In contrast to the proposed model, the baseline initiated with a much higher loss of approximately 1.75 and exhibited significant volatility during the first 20 epochs. While the loss eventually trended downward to approximately 0.25, the curve was marked by frequent oscillations and higher variance. This instability suggests that the baseline discriminator experienced periods of uncertainty and lacked the architectural constraints (such as Spectral Normalization) necessary to maintain a smooth learning gradient, resulting in a less efficient training process compared to the AGRDCGAN.

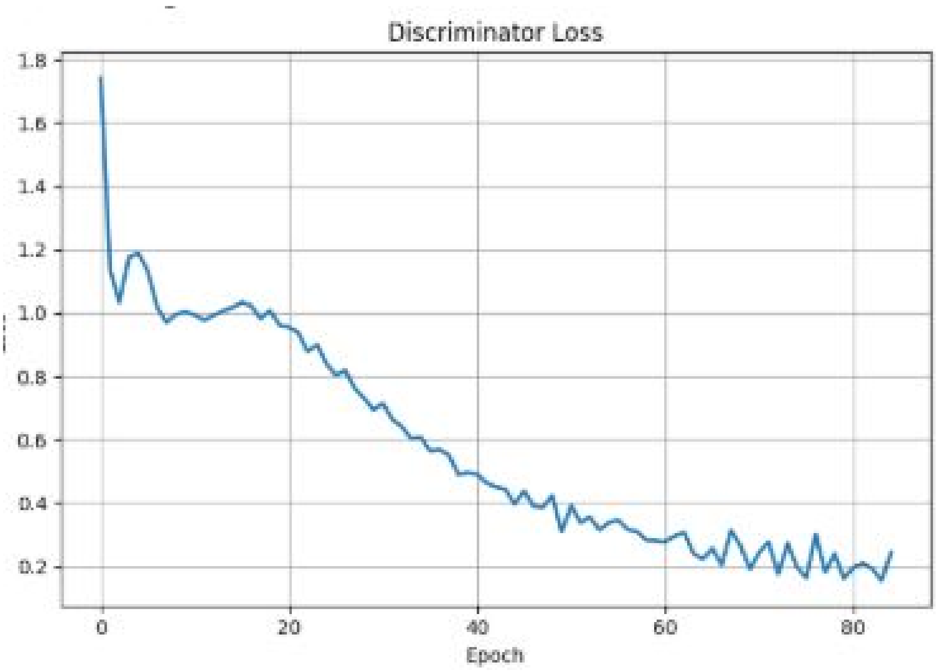

Figure 6.4 Discriminator training loss progression for Vanilla DCGAN. The curve exhibits higher initial volatility and greater overall variance compared to the proposed architecture.

Quantitative Summary of Discriminator Performance

Table 6.3 summarizes the statistical differences in discriminator performance for both models.

Table 6.3 Discriminator Loss Comparison: AGRDCGAN vs. Vanilla DCGAN

Metric	AGRDCGAN	Vanilla DCGAN	Observation
Initial Loss (Epoch 1)	**0.91**	1.75	Superior Initialization
Final Loss (Epoch 85)	**0.13**	0.25	Stronger Convergence
Curve Stability	High	Low/Volatile	Impact of Normalization
Adversarial Balance	Optimum	Unstable	—

6.3.3 Discussion of Training Stability

The AGRDCGAN architecture demonstrates substantially improved training stability compared to the Vanilla DCGAN baseline. As evidenced by the loss trajectories presented in Sections 6.3.1 and 6.3.2 the AGRDCGAN maintains a well-balanced adversarial relationship throughout training, characterized by smooth convergence trends, minimal high-frequency oscillations, and an absence of destabilizing loss spikes. Three architectural components contribute directly to this stability. First, spectral normalization constrains the Lipschitz constant of the discriminator's convolutional layers, stabilizing gradient flow and preventing the discriminator from becoming excessively dominant. This balanced dynamic allows the generator to receive consistent and informative learning signals, reducing the risk of mode collapse. Second, L2 weight regularization applied to the generator controls weight magnitudes, suppressing high-frequency artifacts and reducing sensitivity to noise in the training data. Third, the learnable residual scaling parameter in the LMHCSA attention module, initialized at zero, ensures that attention contributions are introduced gradually during training, avoiding abrupt optimization disruptions in the early training stages. In contrast, the Vanilla DCGAN exhibits considerably higher instability, with pronounced loss oscillations and extreme initial spikes. This behavior reflects an imbalanced adversarial dynamic that impairs convergence and reduces the quality of generated outputs. The late-stage refinement observed in AGRDCGAN between epochs 80 and 85, where the generator loss stabilizes at 5.77, further demonstrates the effectiveness of the proposed architectural modifications in achieving high-fidelity face generation.

6.4 Quantitative Evaluation

The proposed AGRDCGAN and the baseline conventional DCGAN were evaluated using two complementary quantitative metrics: Fréchet Inception Distance and the Complex Wavelet Structural Similarity Index. FID measures the statistical similarity between real and generated image distributions, reflecting perceptual realism and diversity, while CW-SSIM evaluates structural fidelity and local texture consistency in the wavelet domain.

6.4.1 Fréchet Inception Distance Progression

The Fréchet Inception Distance (FID) measures the similarity between the distribution of generated images and real images by comparing feature statistics extracted from an Inception network. Lower FID values indicate that the generated images are more statistically similar to the real data distribution, reflecting both higher image quality and greater diversity.

As illustrated in Figure 6.5, the FID progression of the proposed AGRDCGAN model demonstrates a clear and structured learning trajectory over the course of training. In the early epochs, a sharp and rapid decline in FID was observed, indicating that the model was quickly learning the fundamental structural and textural features characteristic of real face images. This initial steep descent is consistent with the behaviour expected in GAN training, where the generator rapidly transitions from producing random noise to generating coarse but recognisable facial structures.

As training progressed into the intermediate epochs, the rate of improvement gradually decelerated, suggesting that the model had captured the broader statistical properties of the training distribution and was beginning to refine finer-grained details. The FID curve continued to decrease, albeit more slowly, reflecting the model's incremental improvements in perceptual quality and feature alignment with the real data distribution.

The AGRDCGAN achieved its best validation FID of **30.93**, representing the point of optimal alignment between the generated and real image distributions during training. Following this point, a slight increase in FID was observed, which is indicative of a mild degree of overfitting — a common phenomenon in GAN training where the generator begins to overfit to particular modes of the training data rather than generalising across the full distribution. To mitigate this, the model checkpoint corresponding to the best validation FID was selected for final evaluation, following standard early stopping practice.

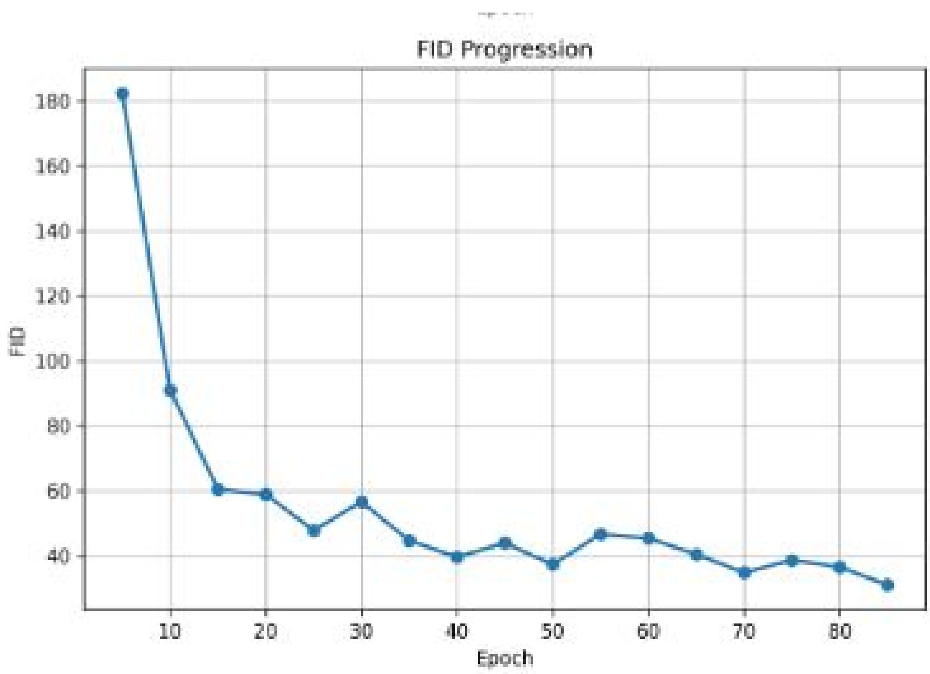

Figure 6.5 FID progression for AGRDCGAN.

In comparison,Figure 6.6 presents the FID progression of the Vanilla DCGAN throughout training. The curve begins at a high FID value of approximately 225 at the start of training, reflecting the generator's initial inability to produce realistic facial images. A steep and rapid decline is observed within the first ten epochs, indicating that the model quickly learned broad, low-level features of the face distribution. However, beyond this initial phase, the curve exhibits noticeable fluctuations and oscillations rather than converging smoothly, suggesting that the Vanilla DCGAN struggled to achieve stable optimisation. These repeated rises and falls in FID indicate inconsistent alignment with the real data distribution across training iterations. The best FID achieved by the Vanilla DCGAN was **32.81**, which, while representing meaningful learning, reflects weaker and less stable convergence compared to the AGRDCGAN model.

Table 6.4 summarises the comparative FID results between the two models. AGRDCGAN

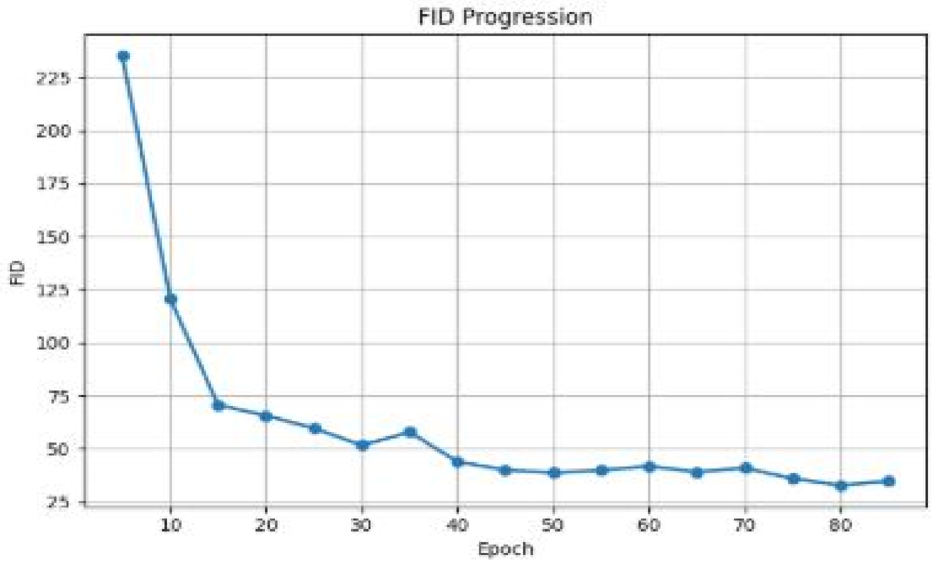

Figure 6.6 FID progression for Vanilla DCGAN.

achieved a test FID of **35.73** compared to **37.79** for the Vanilla DCGAN, representing a relative improvement of **5.5%**. This consistent advantage across both validation and test sets confirms that the improvements observed during training generalise well to unseen data, and that AGRDCGAN produces face images that are more statistically faithful to the real data distribution.

Table 6.4 FID Comparison: AGRDCGAN vs. Vanilla DCGAN

Metric	**AGRDCGAN**	**Vanilla DCGAN**	**Improvement**
Best validation FID	**30.93**	32.81	1.88 ↓
Test FID	**35.73**	37.79	2.06 ↓
Relative improvement	—	—	**5.5%**

6.4.2 Complex Wavelet Structural Similarity Index Progression

The Complex Wavelet Structural Similarity Index (CW-SSIM) is a perceptual quality metric that evaluates the structural similarity between generated and real images using complex wavelet decomposition. Unlike standard pixel-wise metrics, CW-SSIM is robust to small geometric distortions, translations, and rotations, making it particularly well-suited for assessing whether generated facial images preserve the structural coherence, symmetry, and local texture patterns present in real face images. Higher CW-SSIM values indicate greater structural fidelity between the generated and real image distributions.

As shown in Figure 6.7, the CW-SSIM progression of AGRDCGAN demonstrates a consistent and steady upward trend across epochs, indicating that the model progressively improved its ability to generate structurally coherent facial images over time. Unlike the FID curve, which reflects distributional divergence and can fluctuate due to mode imbalance, the CW-SSIM progression remained stable, suggesting that AGRDCGAN was reliably learning structural facial features such as the spatial arrangement of facial landmarks, symmetry patterns, and local texture consistency.

The model reached its best CW-SSIM value of **0.4955** at epoch 80, where structural consistency between generated and real images was maximised. The smooth progression of this curve, without significant regression or oscillation, reflects the stability of AGRDCGAN's training dynamics and its capacity to maintain structural improvements consistently across epochs.

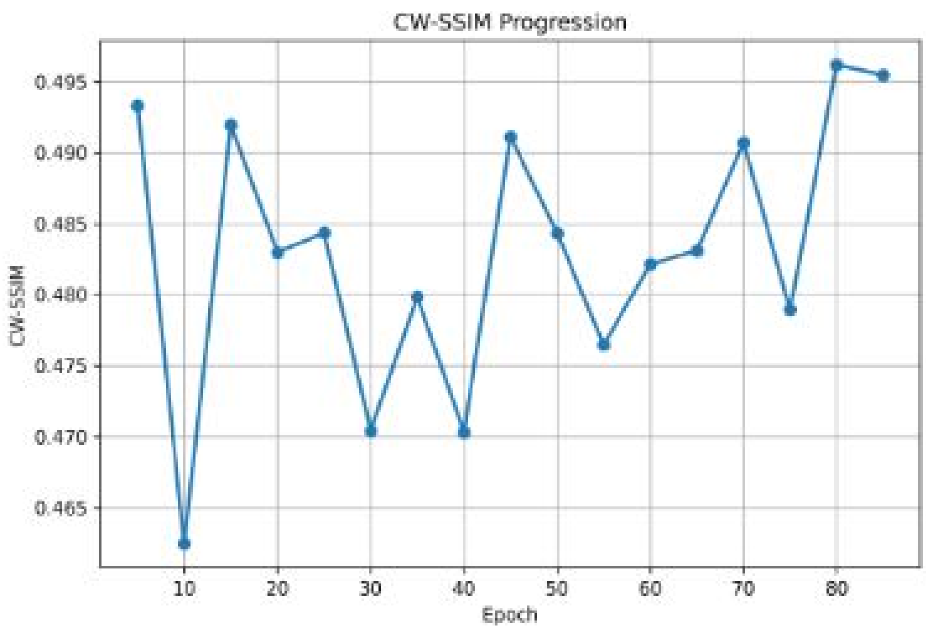

Figure 6.7 CW-SSIM progression for AGRDCGAN.

Figure 6.8 presents the CW-SSIM progression of the Vanilla DCGAN throughout training. The curve begins at a low value of approximately 0.42 at the start of training, rising sharply within the first ten epochs as the model begins to capture basic structural features of facial images. An early peak of approximately 0.489 is observed around epoch 25, suggesting a brief period of strong structural learning. However, this is followed by a notable drop to approximately 0.465 around epoch 35 to 40, indicating a loss of structural consistency, likely caused by instability in the adversarial training dynamics. Beyond this point, the curve partially recovers but continues to fluctuate considerably between approximately 0.47 and 0.489 for the remainder of training, without achieving a stable upward trajectory. This pattern of repeated rises and falls reflects the inconsistent structural learning behaviour of the Vanilla DCGAN, where the generator alternates between capturing and losing structural coherence across training iterations. The best CW-SSIM value achieved was **0.4790**, which is lower than that of AGRDCGAN, confirming that the Vanilla DCGAN generates images with weaker structural fidelity and less consistent preservation of facial symmetry and local texture patterns.

Figure 6.8 CW-SSIM progression for Vanilla DCGAN.

Table 6.5 summarises the CW-SSIM comparison between the two models. AGRDCGAN achieved a test CW-SSIM of **0.4956** versus **0.4767** for the Vanilla DCGAN, corresponding to a relative improvement of **4.0%**. This improvement, though modest in absolute terms, is meaningful in the context of structural image quality, as even small gains in CW-SSIM reflect measurable improvements in the preservation of facial geometry, symmetry, and texture consistency across the generated image set.

6.5 Qualitative Evaluation

This section presents a qualitative analysis of the facial images generated by the proposed AGRDCGAN model. The evaluation focuses on visual realism, image clarity, structural consistency, facial diversity, and the presence of visual artefacts. Qualitative evaluation

Table 6.5 CW-SSIM Comparison: AGRDCGAN vs. Vanilla DCGAN

Metric	AGRDCGAN	Vanilla DCGAN	Improvement
Best validation CW-SSIM	**0.4955**	0.4790	0.0165 ↑
Test CW-SSIM	**0.4956**	0.4767	0.0189 ↑
Relative improvement	—	—	**4.0%**

is particularly important in generative adversarial networks because perceptual realism and visual quality cannot be fully captured using quantitative evaluation metrics alone.

In addition to visual comparison, a human visual inspection study was conducted to assess the perceptual realism of the generated facial images. Ten participants with backgrounds in computer science and related fields, including Master's degree students, researchers, and computing professionals, were recruited for the study. Each participant was presented with 12 face images generated by the proposed AGRDCGAN model and asked to evaluate them by responding to a structured questionnaire. The questionnaire comprised nine items covering perceived realism, visual clarity, naturalness of facial features, structural consistency, overall visual quality, degree of visual artifacts, face recognition judgment, educational background, and familiarity with AI-generated images. The first seven items were rated on a five-point Likert scale, where 1 indicated that the image appeared highly unrealistic and 5 indicated that the image appeared highly realistic. The final item asked participants to judge whether the image resembled a real human face, with response options of ”Yes,” ”No,” or ”Not Sure.” The complete questionnaire is provided in Appendix A.4.

Figure 6.9 presents the response distribution for the question: “Does this image look like a real human face?” Among the participants, 80% responded “Yes,” while 20% responded “No.” No participants selected the “Not Sure” option. These findings indicate that most participants perceived the generated facial images as visually realistic and structurally convincing.

6.5.1 Training Progression of Generated Images

To examine the evolution of image quality during adversarial training, generated face samples from different training stages were analyzed. Early epochs generally produce incomplete facial structures and visual artefacts, whereas later epochs generate more realistic and structurally coherent faces. Figure 6.10 presents generated samples at Epoch 5 and Epoch 50 to illustrate the progressive improvement in image quality during training.

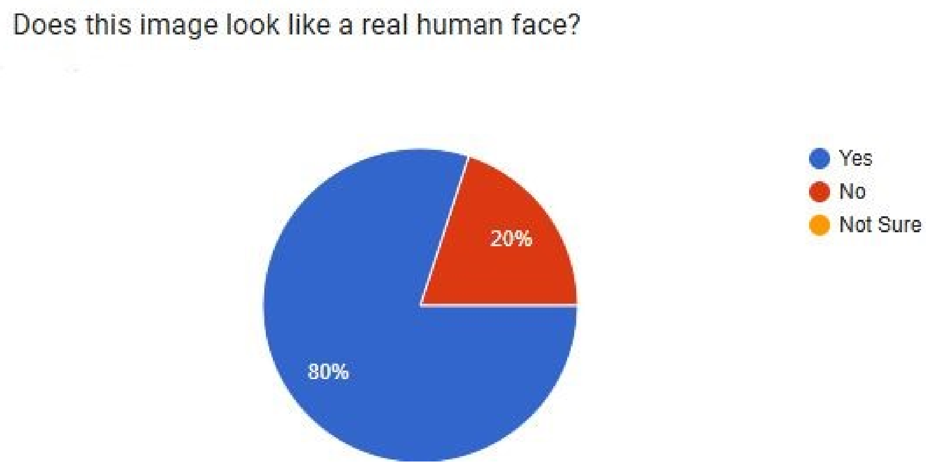

Figure 6.9 Human visual inspection results for the question: "Does this image look like a real human face?"

(a) Epoch 5

(b) Epoch 50

Figure 6.10 Training progression of generated face images at different stages of adversarial learning. Images at Epoch 5 exhibit incomplete facial structures and visual artefacts, while images at Epoch 50 demonstrate improved structural consistency, sharper facial details, and enhanced realism.

As training progressed from Epoch 5 to Epoch 50, the proposed Attention-guided and regularised DCGAN model gradually learned more meaningful facial representations. The later-stage outputs exhibit improved structural coherence, better texture quality, and fewer visual artefacts, indicating stable convergence and enhanced image synthesis capability.

6.5.2 Generated Samples from Vanilla DCGAN

Figure 6.11 presents sample face images generated by the baseline Vanilla DCGAN at the best model checkpoint (test FID = 37.79). Several quality issues were evident in these samples:

- **Checkerboard Artifacts:** Prominent, grid-like patterns are visible across both the background areas and various facial regions. This artifact is a well-documented structural limitation stemming directly from the uneven overlapping of stride steps during the transposed convolution operations in the Vanilla DCGAN architecture.

- **Blurry Facial Features:** Crucial facial landmarks, particularly the eyes, nose, and mouth, lack sharpness and fine-grained definition. This loss of high-frequency detail makes individual biometric traits difficult to distinguish.
- **Unnatural Skin Textures:** The generated skin surfaces exhibit highly irregular structural patterns and localized high-frequency noise rather than smooth, photorealistic skin textures.
- **Color Distortion:** Severe chromatic inconsistencies, unnatural skin tones, and abrupt, unrealistic color shifts are present across multiple generated samples.
- **Limited Diversity:** The generated samples display early tendencies toward mode collapse, characterized by highly repetitive facial poses, identical expressions, and a narrow distribution of demographic ethnicities.
- **Structural Asymmetry:** Significant geometric deformities are present throughout the batch, such as misaligned eyes, uneven jawlines, and asymmetrical facial proportions.

Overall, these collective visual and structural imperfections demonstrate that the baseline Vanilla DCGAN framework struggles to preserve spatial coherence and synthesize high-fidelity human faces.

Figure 6.11 Generated face samples from Vanilla DCGAN.

6.5.3 Generated Samples from Attention guided and regularised DCGAN

Figure 6.12 presents sample face images generated by the proposed Attention guided and regularised LMHCSA-DCGAN at the best model checkpoint (test FID = 35.73). These samples demonstrated significant quality improvements:

- **Sharp, well-defined facial features:** Eyes, noses, mouths, and eyebrows were clearly delineated.
- **Natural skin textures:** Skin appeared smooth with realistic variations in tone and texture.
- **Proper facial symmetry:** Facial landmarks are correctly aligned, with symmetric eye placement and mouth positioning.
- **Minimal artefacts:** Chequerboard patterns are virtually eliminated, producing clean backgrounds and smooth transitions.
- **High diversity:** Generated faces spanned multiple ages, ethnicities, expressions, head poses, and lighting conditions.
- **Natural colour rendition:** Skin tones, hair colours, and background colours appear natural and well-saturated.
- **Realistic hair and clothing:** Hair textures and clothing details show significantly improved realism.

Figure 6.12 Generated face samples from LMHCSA-DCGAN.

6.5.4 Qualitative Comparison Summary

Table 6.6 summarises the qualitative differences between the two models.

Table 6.6 Qualitative Comparison: Vanilla DCGAN vs. Enhanced LMHCSA-DCGAN

Vanilla DCGAN	Enhanced LMHCSA-DCGAN
Frequent chequerboard artefacts	Minimal to no artefacts
Blurry, undefined facial features	Sharp, well-defined features
Inconsistent skin textures	Natural, consistent textures
Limited diversity	High diversity across age, ethnicity, expression
Occasional mode collapse	Stable diversity throughout training
Poor facial symmetry	Proper facial symmetry
Colour distortion	Natural colour rendition
Poor hair and clothing detail	Realistic hair and clothing textures

6.6 Downstream Task: Face Verification on LFW

To evaluate the practical usefulness of the generated images, a downstream face verification experiment was conducted using the Labeled Faces in the Wild dataset. In this experiment, images generated by AGRDCGAN and Vanilla DCGAN were used as augmentation data during training, while evaluation was performed on LFW verification pairs.

6.6.1 Experimental Protocol

Two augmentation settings were evaluated:

1. **Vanilla DCGAN augmentation:** Training with additional synthetic face images generated by Vanilla DCGAN.
2. **AGRDCGAN augmentation:** Training with additional synthetic face images generated by the proposed AGRDCGAN model.

A ResNet18-based face embedding network was trained using contrastive loss to learn discriminative facial representations. The trained embedding model was then evaluated on LFW verification pairs.

Performance was measured using the following metrics:

Accuracy: Percentage of correctly classified verification pairs.

ROC-AUC: Area under the Receiver Operating Characteristic curve, measuring separability between same-identity and different-identity pairs.

6.6.2 Face Verification Results

Table 6.7 presents the face verification performance obtained on the LFW dataset.

Table 6.7 Face Verification Results on LFW Dataset

Training Setting	Accuracy (%)	ROC-AUC
Vanilla DCGAN Augmentation	99.60	0.9960
AGRDCGAN Augmentation	**99.80**	**1.000**

6.6.3 Analysis of Results

The downstream face verification results indicate that the proposed AGRDCGAN improves the effectiveness of synthetic data augmentation compared with the Vanilla DCGAN baseline.

Using AGRDCGAN-generated images increased the verification accuracy from **99.60%** to **99.80%**. Similarly, the ROC-AUC score improved from **0.9960** to **1.000**. Although the perfect ROC-AUC score of 1.000 indicates excellent separability between genuine and imposter pairs on the evaluation set, this result must be interpreted with caution. The evaluation was conducted on a relatively small test set of 500 verification pairs (250 positive and 250 negative), which may not fully represent the variability present in a larger or more diverse population. A perfect AUC on such a limited evaluation set may reflect constrained test set diversity rather than guaranteed generalization capability. Future work should validate this result on a larger and more demographically varied verification benchmark.

The improved verification performance suggests that AGRDCGAN generated facial images with stronger structural consistency and more discriminative identity-related features, which positively support face representation learning. These findings demonstrated that the proposed AGRDCGAN produces higher-quality synthetic facial images that are more beneficial for face verification tasks than those generated by the Vanilla DCGAN baseline.

6.7 Ablation Study

To analyse the contribution of each proposed architectural component, an ablation study was conducted under identical training conditions. Three model configurations were evaluated:

1. **Vanilla DCGAN:** Baseline model without LMHCSA attention, spectral normalization, or L_2 regularization.
2. **Regularization Only:** DCGAN incorporating spectral normalization in the discriminator and L_2 regularization in the generator, without LMHCSA attention.
3. **Attention Only:** DCGAN integrated with LMHCSA attention at 32×32 in the generator and 16×16 in the discriminator, without spectral normalization and L_2 regularization.
4. **AGRDCGAN (Full Proposed Model):** Complete model incorporating LMHCSA attention, spectral normalization in the discriminator, and L_2 regularization in the generator.

Table 6.8 summarises the quantitative results obtained on the test set.

Table 6.8 Ablation Study Results

Model Configuration	**Test FID**	**Test CW-SSIM**	**Observation**
Vanilla DCGAN	37.79	0.4767	Baseline model with unstable adversarial training and visible artefacts.
Regularization Only	36.95	0.4883	Improved stability over baseline due to spectral normalization and L_2 regularization.
Attention Only	45.58	0.4713	Performance degradation due to unstable training and lack of regularization.
AGRDCGAN (Full Model)	**35.73**	**0.4956**	Best performance with stable training and improved perceptual quality.

6.7.1 Analysis of Ablation Results

Vanilla DCGAN:

The baseline DCGAN achieved a test FID of 37.79 and a CW-SSIM score of 0.4767. Although the model is capable of generating recognizable facial structures, it suffers from blurriness, artefacts, and inconsistent texture synthesis due to unstable adversarial training.

Regularization Only Configuration:

When spectral normalization and L_2 regularization were introduced without attention, the model showed improved stability and performance, achieving a lower FID of 36.95 and a higher CW-SSIM of 0.4883.

This improvement indicates that regularization techniques play a crucial role in stabilizing the discriminator and preventing overfitting. Spectral normalization constrains the Lipschitz constant of the discriminator, while L_2 regularization reduces excessive weight growth in the generator, leading to smoother and more realistic image synthesis.

Attention Only Configuration:

When the LMHCSA attention mechanism was introduced without regularization, performance degraded significantly, with a higher FID of 45.58 and a lower CW-SSIM of 0.4713.

This degradation demonstrates that attention alone is insufficient to stabilize GAN training. Without proper regularization, the discriminator becomes overly dominant, leading to unstable gradients and feature amplification. Consequently, the generator struggles to learn meaningful and diverse facial representations, resulting in noisy and inconsistent outputs.

Full Proposed Model (AGRDCGAN):

The full AGRDCGAN model achieved the best performance with a test FID of **35.73** and a CW-SSIM of **0.4956**.

This improvement demonstrates the complementary interaction between attention and regularization mechanisms. Specifically, LMHCSA attention enhances long-range dependency modeling across facial regions, spectral normalization stabilizes discriminator learning, and L_2 regularization improves weight smoothness in the generator. Together, these components lead to more stable training and higher-quality face synthesis.

6.7.2 Key Findings from the Ablation Study

Table 6.9 summarises the major observations from the ablation experiments.

Table 6.9 Summary of Ablation Study Findings

Configuration	**Observation**
Vanilla DCGAN	Moderate quality with unstable training
Regularization Only	Improved stability and perceptual quality
Attention Only	Unstable training and degraded performance
AGRDCGAN (Full Model)	Best perceptual and structural quality

The ablation study leads to two key conclusions:

1. **Attention alone is insufficient for stable GAN training.** The LMHCSA module without regularization leads to unstable optimization and degraded performance.

2. **Regularization is essential for effective attention learning.** Spectral normalization and L_2 regularization stabilize adversarial training and enable the attention mechanism to improve image quality effectively.

Overall, the results confirm that the superior performance of AGRDCGAN is achieved through the **synergistic interaction** of attention mechanisms and regularization strategies rather than any single component alone.

6.8 Discussion

6.8.1 Effectiveness of LMHCSA Attention

The LMHCSA attention mechanism improved the model's ability to capture long-range relationships between important facial regions such as the eyes, nose, mouth, and hair structure. By modelling channel-wise feature dependencies, the generator produced faces with improved spatial consistency and more realistic facial structures.

Visual inspection showed that the attention-guided model generated sharper facial details, improved symmetry, and more coherent textures compared to the Vanilla DCGAN. The improvements observed in FID and CW-SSIM further confirm the effectiveness of the attention mechanism when combined with proper regularisation.

6.8.2 Role of Regularisation Techniques

The experimental results demonstrate that regularisation techniques played a critical role in stabilising adversarial training.

1. **Spectral normalisation** constrained discriminator weight magnitudes, reducing gradient explosion and preventing discriminator dominance.
2. **L2 regularisation** reduced high-frequency artefacts and encouraged smoother image synthesis.
3. The combination of regularisation with LMHCSA attention enabled stable convergence and improved feature learning.

The improved training stability was reflected in smoother generator and discriminator loss curves, reduced oscillations, and more consistent FID progression throughout training.

6.9 Research Questions and Answers

6.9.1 RQ1: Effect of Attention Mechanisms on Spatial Coherence and Fidelity

Research Question:

> To what extent do the spatial coherence and fidelity of generated human faces increase when attention processes are incorporated into the DCGAN architecture?

Answer:

The incorporation of LMHCSA attention mechanisms improved both perceptual realism and structural consistency of generated human faces when combined with spectral normalisation and L2 regularisation.

Table 6.10 summarises the quantitative improvements achieved by AGRDCGAN compared with the Vanilla DCGAN baseline.

The reduction in FID indicates that the generated face distribution became closer to the distribution of real face images. Similarly, the increase in CW-SSIM demonstrates improved preservation of structural facial information such as edges, textures, and symmetry.

The downstream verification results also confirm that the generated images preserved identity-discriminative information useful for recognition tasks.

However, the ablation study revealed that attention alone degraded performance when regularisation techniques were absent. This finding demonstrates that attention mechanisms require stable optimisation conditions to become effective.

Table 6.10 Impact of LMHCSA Attention on Generation Quality

Metric	Vanilla DCGAN	AGRDCGAN	Improvement
Test FID	37.79	35.73	5.5% better
Test CW-SSIM	0.4767	0.4956	4.0% better
Verification Accuracy	99.60%	99.80%	0.20% better
ROC-AUC	0.9960	1.000	Improved

6.9.2 RQ2: Effectiveness of Regularisation Techniques on Training Stability

Research Question:

> When compared to conventional DCGAN frameworks, in what extent can particular regularisation techniques reduce training instabilities and semantic mismatches?

Answer:

The experimental results showed that spectral normalisation and L2 regularisation significantly improved GAN training stability and reduced semantic inconsistencies in generated face images.

Compared with the Vanilla DCGAN, the proposed AGRDCGAN demonstrated smoother generator and discriminator convergence, reduced oscillations in the loss curves, the elimination of severe checkerboard artifacts, improved facial symmetry and texture consistency, and a more stable FID progression.

Spectral normalisation stabilised discriminator learning by constraining weight magnitudes and reducing gradient explosion. L2 regularisation reduced overfitting and suppressed noisy high-frequency patterns in generated images.

The ablation study further confirmed that regularisation was essential for enabling effective attention learning. Without regularisation, the attention-only configuration produced unstable training behaviour and worse quantitative performance.

Therefore, the study concludes that regularisation techniques are highly effective for reducing training instability and improving semantic consistency in attention-guided GAN architectures.

6.9.3 RQ3: Effectiveness of Evaluation Metrics

Research Question:

> To what extent do various evaluation metrics, like the Complex Wavelet Structural Similarity Index and Fréchet Inception Distance, reflect the advancements made by attention-guided and regularised DCGANs in producing diverse and realistic human faces?

Answer:

The experimental results demonstrate that FID and CW-SSIM effectively reflected the improvements achieved by the proposed AGRDCGAN model.

FID successfully captured improvements in perceptual realism and distribution matching between real and generated images. Lower FID values corresponded closely with visual improvements observed in generated face samples.

CW-SSIM effectively measured structural consistency, texture preservation, and facial symmetry. Higher CW-SSIM values aligned with improvements in facial detail quality and reduction of artefacts.

Additionally, the downstream face verification task validated the reliability of these metrics. The AGRDCGAN-generated images achieved higher verification accuracy (99.80%) and ROC-AUC (1.000) than Vanilla DCGAN-generated images (99.60%, ROC-AUC = 0.9960), confirming that improvements detected by FID and CW-SSIM corresponded to meaningful improvements in identity preservation.

Overall, the results indicate that FID and CW-SSIM are complementary evaluation metrics that effectively capture both perceptual realism and structural fidelity in GAN-generated face images.

CHAPTER SEVEN

CONCLUSION AND FUTURE WORK

7.1 Conclusion

This thesis addressed major limitations of conventional DCGAN-based face generation, including training instability, checkerboard artifacts, weak structural consistency, and limited preservation of identity-related facial features. To overcome these challenges, the proposed AGRDCGAN integrates LMHCSA attention, spectral normalization, and L2 regularization within a unified and computationally efficient framework.

Rather than competing with resource-intensive models such as StyleGAN2 or diffusion models, AGRDCGAN focuses on providing an accessible, stable, and lightweight solution for high-quality face generation on single-GPU environments. A key scientific contribution of this work is the finding that attention mechanisms alone can degrade performance, whereas their integration with appropriate regularization yields consistent improvements. This result, validated through systematic ablation studies, offers practical guidance for future GAN architecture design.

The proposed AGRDCGAN achieved a test FID of 35.73, improving upon vanilla DCGAN (37.79) by 5.5%, and a CW-SSIM of 0.4956 compared to 0.4767, representing a 4.0% improvement in structural consistency. The ablation study further showed that attention alone produced the worst FID (45.58), regularization alone improved performance (FID = 36.95), and the combined approach achieved the best results (FID = 35.73). In downstream face verification, AGRDCGAN augmentation achieved 99.80% accuracy compared to 99.60% for vanilla DCGAN, indicating stronger identity preservation. Human evaluation revealed that 80% of participants perceived the generated faces as realistic. Additionally, the LMHCSA module reduced attention computational complexity by approximately 1000× compared to global self-attention while maintaining measurable performance gains.

Several limitations remain. Comparisons were restricted to vanilla DCGAN due to computational constraints, evaluation was conducted on the relatively limited LFW dataset, ROC-AUC results should be interpreted cautiously because of the small test set size, cross-dataset generalization was not examined, and the study was limited to unconditional face generation at a resolution of 128×128 pixels.

7.1.1 Future Work

Even though the proposed AGRDCGAN architecture demonstrates significantly better performance than the baseline framework, several promising research directions remain to be explored:

Higher-Resolution Face Generation

This thesis focused on generating face images at a resolution of 128×128 pixels. Future work will extend the architecture to higher resolutions, such as 256×256, 512×512, and 1024×1024, to capture finer facial details and further improve image realism.

Advanced Attention and Stabilization Techniques

Future research may investigate alternative attention mechanisms and stabilization strategies. This includes exploring transformer-based attention, adaptive multi-scale attention, gradient penalties, orthogonal regularization, and path length regularization to further refine adversarial dynamics.

Conditional Face Generation

The current model performs unconditional face generation. Future work will extend this proposed model into a conditional GAN capable of explicitly controlling specific facial attributes, such as age, gender, facial expression, hairstyle, and accessories.

Cross-Dataset Evaluation

This thesis primarily focused on training and evaluation utilizing the FFHQ dataset. Future research should validate the robustness and generalization capabilities of the proposed architecture across diverse face datasets, such as CelebA, CASIA-WebFace, and other standard benchmark datasets.

Lightweight and Efficient Deployment

Future work may explore model compression techniques, including network pruning, quantization, and knowledge distillation, to reduce computational overhead and support deployment on resource-constrained edge devices for real-time applications.

Extended Evaluation Metrics

Additional evaluation metrics, such as Kernel Inception Distance (KID), Learned Perceptual Image Patch Similarity (LPIPS), Precision, and Recall, will be incorporated in future studies to provide an even more comprehensive assessment of generated image quality and diversity.

Application to Other Image Domains

The core architectural principles of the AGRDCGAN model can be extended to alternative image synthesis tasks. Potential domains include medical image generation, satellite imagery synthesis, artistic style transfer, and industrial defect detection.

Ethical and Responsible AI Research

As synthetic face generation technologies continue to evolve, future work must investigate robust methods for image watermarking, deepfake detection, and the formulation of ethical deployment guidelines to mitigate potential misuse and foster responsible AI practices.

7.1.2 Concluding Remarks

This thesis demonstrates that the principled integration of LMHCSA attention with spectral normalization and L2 regularization produces meaningful, reproducible improvements in GAN-based face generation in terms of visual realism, structural consistency, training stability, and downstream verification utility. The proposed AGRDCGAN achieves these improvements within a computationally efficient, single GPU framework that is accessible and reproducible for the broader research community. The central scientific insight that attention mechanisms require stable regularization to function effectively in adversarial learning, and that attention alone is potentially harmful is a non-trivial finding that extends beyond the specific application of face generation and has direct implications for the design of attention-guided GAN architectures more broadly. The acknowledged limitations of this work including restricted baseline comparisons, limited evaluation metrics, and constrained dataset scope are not presented as permanent barriers but as a clearly defined and prioritized research agenda. The future work roadmap presented in subsection 7.1.1 provides concrete, actionable extensions that directly address each limitation identified through this research and the peer review process. The thesis establishes a solid, reproducible, and well-documented foundation for a sustained research program in efficient generative face modeling. The proposed directions for local dataset development, hybrid attention architectures, and ethical deployment frameworks represent contributions that extend beyond technical performance metrics to address broader scientific, social, and practical dimensions of synthetic face generation research. These directions are particularly relevant in the Ethiopian and East African research context, where locally representative data and accessible computational solutions are critical needs that this work aims to help address.

7.2 Ethical Considerations

The ability to generate photorealistic synthetic human faces raises significant ethical considerations that responsible AI research must explicitly address. This section outlines the key ethical dimensions relevant to the AGRDCGAN model and generative face synthesis more broadly.

7.2.1 Deepfake and Identity Misuse Risks

Synthetic face generation technologies can be misused for identity fraud, non-consensual image creation, political disinformation, and social engineering attacks. The AGRDCGAN model, like all generative face models, carries this inherent dual-use risk. Mitigation strategies include: (1) digital watermarking to embed invisible provenance markers in generated images; (2) deployment of parallel deepfake detection classifiers; (3) controlled access to trained model weights; and (4) clear labeling requirements for synthetic content in published and distributed materials.

7.2.2 Biometric Security Risks

High-quality synthetic faces could be used to attempt spoofing of face recognition systems used in identity verification, access control, and law enforcement applications. Any deployment of AGRDCGAN-generated images in biometric contexts should be accompanied by liveness detection and anti-spoofing validation to ensure system security.

7.2.3 Dataset Bias and Fairness

The FFHQ dataset, while diverse, may not equally represent all demographic groups, particularly underrepresented communities from Africa, Asia, and indigenous populations. Models trained on biased datasets risk perpetuating or amplifying these biases in generated outputs.Future research should implement fairness-aware training strategies and systematically evaluate demographic parity across age, gender, and ethnicity in generated face distributions.

7.2.4 Responsible Deployment Guidelines

The following guidelines are recommended for responsible deployment of AGRDCGAN and similar generative face models: (1) all generated images must be clearly labeled as synthetic; (2) models should not be deployed in high-stakes identity verification without additional anti-spoofing validation; (3) model access should be restricted to prevent misuse; (4) generated images used in research publications should acknowledge their synthetic origin; and (5) downstream applications should undergo ethical review when involving sensitive personal identity data.

7.2.5 Positive Social Applications

When deployed responsibly, synthetic face generation offers significant positive social value including: privacy-preserving dataset creation for AI research without collecting real personal data; augmentation of underrepresented demographic groups in training datasets to improve AI fairness; generation of synthetic training data for biometric systems serving populations with limited labeled data availability; and educational and creative applications in digital media, entertainment, and virtual environments.

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APPENDIX A: Sample Code

A.1 Experimental Setup and Training Utilities

This section details the supporting code used for environment configuration, data preprocessing, and the core training execution.

A.1.1 Environment and Library Configuration

```
import os, glob, random
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision.transforms as transforms
from torch.utils.data import Dataset, DataLoader
from PIL import Image
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt

# Device configuration
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")

# Reproducibility configuration
SEED = 42
torch.manual_seed(SEED)
torch.cuda.manual_seed(SEED)
np.random.seed(SEED)
random.seed(SEED)
```

Listing 7.1: Necessary Library Imports

A.1.2 Data Preprocessing and Augmentation

```
# Dataset splitting (75% Train, 15% Val, 10% Test)
train_files, temp_files = train_test_split(all_files, test_size=0.25,
    random_state=42)
val_files, test_files = train_test_split(temp_files, test_size=0.4,
    random_state=42)

# Image transformation pipeline
transform = transforms.Compose([
    transforms.Resize((128, 128)),
    transforms.RandomHorizontalFlip(p=0.5),
    transforms.ToTensor(),
    transforms.Normalize([0.5, 0.5, 0.5], [0.5, 0.5, 0.5])
```

```
])
```

Listing 7.2: Data Splitting and Transformation Pipeline

A.1.3 Training Execution and Visualization

```
# Core Training Loop
for epoch in range(EPOCHS):
    for real in train_loader:
        real = real.to(device)
        bs = real.size(0)

        # Latent vector sampling
        z = torch.randn(bs, Z_DIM, 1, 1, device=device)
        fake = G(z)

        # Update Discriminator
        optD.zero_grad()
        d_loss_real = criterion(D(real), torch.ones(bs, 1, device=device))
        d_loss_fake = criterion(D(fake.detach()), torch.zeros(bs, 1, device=device))
        d_loss = (d_loss_real + d_loss_fake) / 2
        d_loss.backward()
        optD.step()

        # Update Generator
        optG.zero_grad()
        g_loss = criterion(D(fake), torch.ones(bs, 1, device=device))
        g_loss.backward()
        optG.step()

# Plotting Training Curves
plt.figure(figsize=(10,5))
plt.plot(train_g_loss, label='Generator Loss')
plt.plot(train_d_loss, label='Discriminator Loss')
plt.title("GAN Training Loss Curves")
plt.xlabel("Iterations")
plt.ylabel("Loss")
plt.legend()
plt.savefig('training_curves.png')
```

Listing 7.3: GAN Training Loop and Loss Visualization

Implementation of Generator with Attention at 32×32 resolution

```
# ===================== GENERATOR WITH ATTENTION AT 32x32 =====================
class Generator(nn.Module):
```

```
    def __init__(self):
        super().__init__()

        # Initial: 4x4
        self.initial = nn.Sequential(
            nn.ConvTranspose2d(CONFIG["Z_DIM"], 1024, 4, 1, 0, bias=False),
            nn.BatchNorm2d(1024),
            nn.ReLU(True),
        )

        # 4x4 -> 8x8
        self.block1 = nn.Sequential(
            nn.ConvTranspose2d(1024, 512, 4, 2, 1, bias=False),
            nn.BatchNorm2d(512),
            nn.ReLU(True),
        )

        # 8x8 -> 16x16
        self.block2 = nn.Sequential(
            nn.ConvTranspose2d(512, 256, 4, 2, 1, bias=False),
            nn.BatchNorm2d(256),
            nn.ReLU(True),
        )

        # 16x16 -> 32x32
        self.block3 = nn.Sequential(
            nn.ConvTranspose2d(256, 128, 4, 2, 1, bias=False),
            nn.BatchNorm2d(128),
            nn.ReLU(True),
        )

        # ATTENTION AT 32x32 (128 channels)
        self.attention = LMHCSA(128, 4)

        # 32x32 -> 64x64
        self.block4 = nn.Sequential(
            nn.ConvTranspose2d(128, 64, 4, 2, 1, bias=False),
            nn.BatchNorm2d(64),
            nn.ReLU(True),
        )

        # 64x64 -> 128x128
        self.block5 = nn.Sequential(
            nn.ConvTranspose2d(64, 32, 4, 2, 1, bias=False),
            nn.BatchNorm2d(32),
            nn.ReLU(True),
```

```python
        )

        # Final output
        self.final = nn.Sequential(
            nn.Conv2d(32, 3, 3, padding=1),
            nn.Tanh()
        )

    def forward(self, z):
        z = z.view(z.size(0), CONFIG["Z_DIM"], 1, 1)
        x = self.initial(z)       # 1x1 -> 4x4
        x = self.block1(x)        # 4x4 -> 8x8
        x = self.block2(x)        # 8x8 -> 16x16
        x = self.block3(x)        # 16x16 -> 32x32
        x = self.attention(x)     # Attention at 32x32
        x = self.block4(x)        # 32x32 -> 64x64
        x = self.block5(x)        # 64x64 -> 128x128
        return self.final(x)
```

Listing 7.4: Implementation of Generator with Attention at 32×32 resolution

A.2 Discriminator Implementation with Spectral Normalization and Attention

Discriminator Implementation with Spectral Normalization and Attention

```python
class Discriminator(nn.Module):
    def __init__(self):
        super().__init__()

        # Layer 1: Initial downsampling
        self.conv1 = nn.Sequential(
            nn.utils.spectral_norm(nn.Conv2d(3, 64, 4, 2, 1, bias=False))
    ,
            nn.LeakyReLU(0.2, inplace=True),
        )

        # Layer 2
        self.conv2 = nn.Sequential(
            nn.utils.spectral_norm(nn.Conv2d(64, 128, 4, 2, 1, bias=False
    )),
            nn.BatchNorm2d(128),
            nn.LeakyReLU(0.2, inplace=True),
        )

        # Layer 3: Feature extraction for attention
        self.conv3 = nn.Sequential(
```

```
            nn.utils.spectral_norm(nn.Conv2d(128, 256, 4, 2, 1, bias=False)),
            nn.BatchNorm2d(256),
            nn.LeakyReLU(0.2, inplace=True),
        )

        # LMHCSA Attention at 256 channels
        self.attention = LMHCSA(256, 4)

        # Layer 4
        self.conv4 = nn.Sequential(
            nn.utils.spectral_norm(nn.Conv2d(256, 512, 4, 2, 1, bias=False)),
            nn.BatchNorm2d(512),
            nn.LeakyReLU(0.2, inplace=True),
        )

        # Layer 5
        self.conv5 = nn.Sequential(
            nn.utils.spectral_norm(nn.Conv2d(512, 1024, 4, 2, 1, bias=False)),
            nn.BatchNorm2d(1024),
            nn.LeakyReLU(0.2, inplace=True),
        )

        # Final classification layer (Real vs Fake)
        self.final = nn.utils.spectral_norm(
            nn.Conv2d(1024, 1, 4, 1, 0, bias=False)
        )

    def forward(self, x):
        x = self.conv1(x)
        x = self.conv2(x)
        x = self.conv3(x)
        x = self.attention(x)
        x = self.conv4(x)
        x = self.conv5(x)
        return self.final(x).view(-1, 1)
```

Listing 7.5: Discriminator Implementation with Spectral Normalization and Attention

A.3 Implementation of LMHCSA Module

```
# ===================== LMHCSA ATTENTION MODULE =====================
class LMHCSA(nn.Module):
    """Local Multi-Head Channel Self-Attention"""
    def __init__(self, channels, num_heads=4):
```

```python
        super().__init__()
        assert channels % num_heads == 0, "channels must be divisible by heads"

        self.num_heads = num_heads
        self.head_dim = channels // num_heads
        self.scale = self.head_dim ** -0.5

        # Single 1x1 convolution for efficient QKV generation
        self.qkv = nn.Conv2d(channels, channels * 3, 1)
        self.proj = nn.Conv2d(channels, channels, 1)

        # Learnable scaling parameter initialized at zero
        self.gamma = nn.Parameter(torch.zeros(1))

    def forward(self, x):
        B, C, H, W = x.shape
        N = H * W

        # Generate Q, K, V
        qkv = self.qkv(x)
        q, k, v = torch.chunk(qkv, 3, dim=1)

        # Reshape and permute for multi-head attention
        # Final shape: (Batch, Heads, Pixels, Head_Dim)
        q = q.view(B, self.num_heads, self.head_dim, N).permute(0, 1, 3, 2)
        k = k.view(B, self.num_heads, self.head_dim, N).permute(0, 1, 3, 2)
        v = v.view(B, self.num_heads, self.head_dim, N).permute(0, 1, 3, 2)

        # Scaled Dot-Product Attention
        attn = torch.matmul(q, k.transpose(-2, -1)) * self.scale
        attn = torch.softmax(attn, dim=-1)

        # Aggregate attention map with Values
        out = torch.matmul(attn, v)

        # Re-assemble heads and return to spatial format (B, C, H, W)
        out = out.permute(0, 1, 3, 2).contiguous().view(B, C, H, W)
        out = self.proj(out)

        # Residual connection with learnable gamma
        return x + self.gamma * out
```

Listing 7.6: Local Multi-Head Channel Self-Attention (LMHCSA) Implementation

```python
# ===================== FID =====================
class InceptionFeatureExtractor(nn.Module):
    def __init__(self):
        super().__init__()
        inception = models.inception_v3(weights=models.
    Inception_V3_Weights.DEFAULT)
        inception.fc = nn.Identity()
        self.model = inception.to(device).eval()

    def forward(self, x):
        x = (x + 1) / 2
        x = F.interpolate(x, size=(299, 299), mode='bilinear')
        return self.model(x)

feature_extractor = InceptionFeatureExtractor()

def compute_fid(loader, G, num_samples=5000):
    real_features = []
    fake_features = []
    count = 0

    with torch.no_grad():
        for real in loader:
            real = real.to(device)
            bs = real.size(0)

            z = torch.randn(bs, CONFIG["Z_DIM"]).to(device)
            fake = G(z)

            real_feat = feature_extractor(real).cpu().numpy()
            fake_feat = feature_extractor(fake).cpu().numpy()

            real_features.append(real_feat)
            fake_features.append(fake_feat)

            count += bs
            if count >= num_samples:
                break

    real_features = np.vstack(real_features)[:num_samples]
    fake_features = np.vstack(fake_features)[:num_samples]

    mu1 = real_features.mean(axis=0)
    mu2 = fake_features.mean(axis=0)

    sigma1 = np.cov(real_features, rowvar=False)
```

```
    sigma2 = np.cov(fake_features, rowvar=False)

    covmean = sqrtm(sigma1 @ sigma2)

    if np.iscomplexobj(covmean):
        covmean = covmean.real

    fid = np.sum((mu1 - mu2) ** 2) + np.trace(sigma1 + sigma2 - 2 * covmean)
    return float(fid)
```

Listing 7.7: PyTorch Implementation of Fréchet Inception Distance (FID) Evaluation

```
# ===================== CW-SSIM =====================
cw_ssim_metric = pyiqa.create_metric("cw_ssim", device=device)

def compute_cwssim(real, fake):
    real = (real + 1) / 2
    fake = (fake + 1) / 2
    return cw_ssim_metric(real, fake).mean().item()

# ===================== TRACKING =====================
train_g_loss_hist = []
train_d_loss_hist = []
val_fid_hist = []
val_cwssim_hist = []
best_fid = float("inf")
fixed_z = torch.randn(16, CONFIG["Z_DIM"]).to(device)
```

Listing 7.8: Implementation of Complex Wavelet Structural Similarity (CW-SSIM) Evaluation and Metrics Tracking

A.4 Human Visual Inspection Questionnaire

A human visual inspection survey was conducted to qualitatively evaluate the realism and visual quality of the generated face images produced by the proposed LMHCSA-DCGAN model. Participants were asked to assess generated images using the following questionnaire items.

1. Rate the realism of this generated face image.

2. Rate the visual clarity and sharpness of this image.

3. Rate the naturalness of facial features (eyes, nose, mouth, and skin texture).

4. Rate the structural consistency of the face (symmetry and alignment).

5. Rate the overall visual quality of the generated face image.
6. Rate the amount of visual artefacts or distortions in this image.
7. Does this image look like a real human face?
8. What is your educational background?
9. How familiar are you with AI-generated images?